UPC ϕ Meson Analysis Note

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1 Introduction

In this analysis, we measure coherent and incoherent ϕ meson photoproduction in ultra-peripheral Au+Au collisions (UPCs) at STAR. The ϕ mesons are reconstructed through the charged-kaon decay channel, $\phi \to K^+K^-$. The primary goal of the analysis is to extract the differential production cross sections for the coherent and incoherent components.

Section 2 describes the dataset, trigger, and luminosity, and Section 3 the event selection. Section 4 establishes the track-quality requirement and the kaon identification, and Section 5 the coherent and incoherent candidate selections and their raw yields. Section 6 presents the vertex and BBC-veto efficiencies. Track distributions of the selected candidates are collected in Appendix A.

2 Dataset and Trigger

This analysis uses the Run-19 Au+Au dataset collected at $\sqrt{s_{NN}} = 200$ GeV. Run-19 is the first Au+Au dataset collected after the full installation of the inner Time Projection Chamber (iTPC) upgrade. The improved inner-sector acceptance allows charged tracks to be reconstructed down to transverse momenta of approximately $p_T \sim 60$ MeV/c. This is particularly important for this analysis because the decay kaons are soft. The ϕ carries very little momentum, especially in coherent production, and in the ϕ rest frame each kaon has a momentum

$$p^* = \sqrt{\left(\frac{m_\phi}{2}\right)^2 - m_K^2} \approx 127 \text{ MeV}/c, \tag{1}$$

so in the laboratory frame the kaon transverse momenta are of order 100 MeV/c.

The dataset used in this analysis has the following configuration:

- Production library: `SL23e`;
- Data format: `MuDst`.

No dedicated UPC trigger was available for this dataset. Instead, events are selected from the Zero Degree Calorimeter (ZDC) minimum-bias (zdcmb) trigger sample.

In UPCs of heavy nuclei, virtual photon exchanges can electromagnetically excite one or both nuclei, most commonly through the Giant Dipole Resonance (GDR). The excited nucleus subsequently de-excites through the emission of one or more forward neutrons, which remain close to beam rapidity and are detected by the ZDCs. The zdcmb trigger therefore provides a baseline sample of UPC events accompanied by mutual electromagnetic dissociation.

The trigger configuration is summarized below:

- Offline trigger ID: `700001`;
- Trigger requirements: ZDC-East, ZDC-West, and ZDC TAC coincidence, with no minimum Time-of-Flight (TOF) requirement;
- Number of recorded events: 138.38 million.

The absence of a TOF requirement is essential for this analysis: as discussed above, the decay kaons are largely too soft to reach the TOF, so a trigger requiring TOF hits would remove most of the coherent signal.

The analyzed dataset was recorded with ZDC coincidence rates ranging from 1.5 kHz to 5.0 kHz, as shown in Figure 1. This corresponds to a clean, low-luminosity running condition. At this collision rate, the mean time separation between collisions is 200–700 μs, which is much longer than the TPC drift time of $\sim 40\,\mu s$.

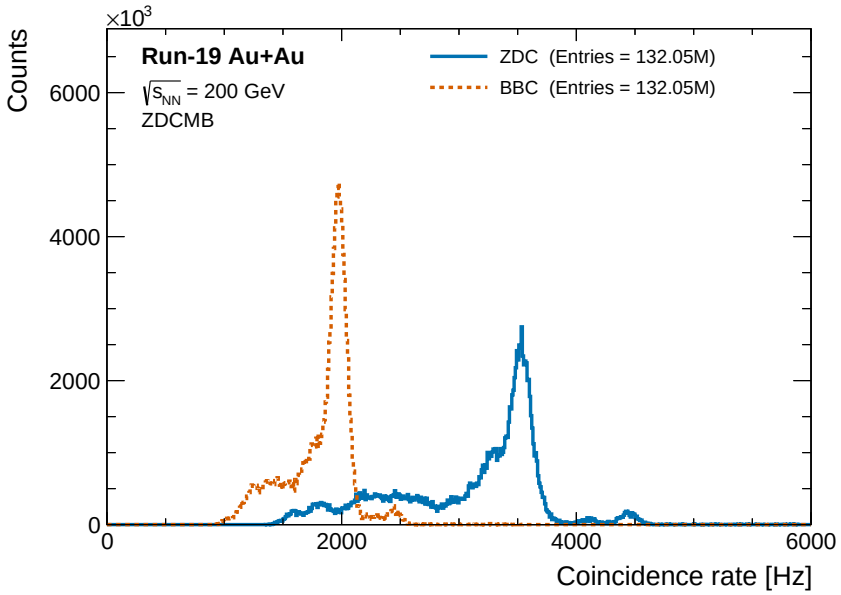


Figure 1: ZDC and BBC coincidence rates for the Run-19 Au+Au zdcmb trigger.

In the Run-19 trigger configuration, there is no cut on the ZDC ADC values, so no ZDC selection efficiency is applied in this analysis. The unattenuated ZDC ADC values that saturate the 12-bit ADC range are placed in the overflow bin, which appears as a spike at the right edge of the ZDC ADC distribution in Figure 2.

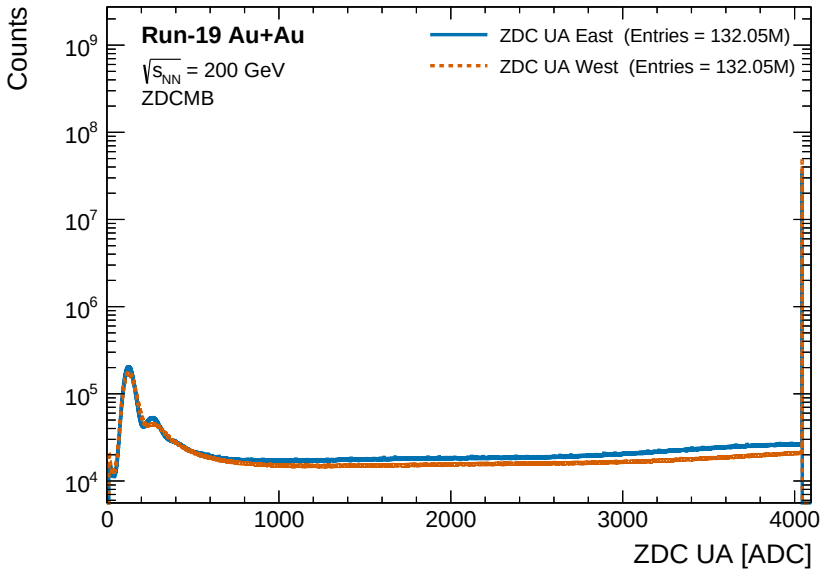
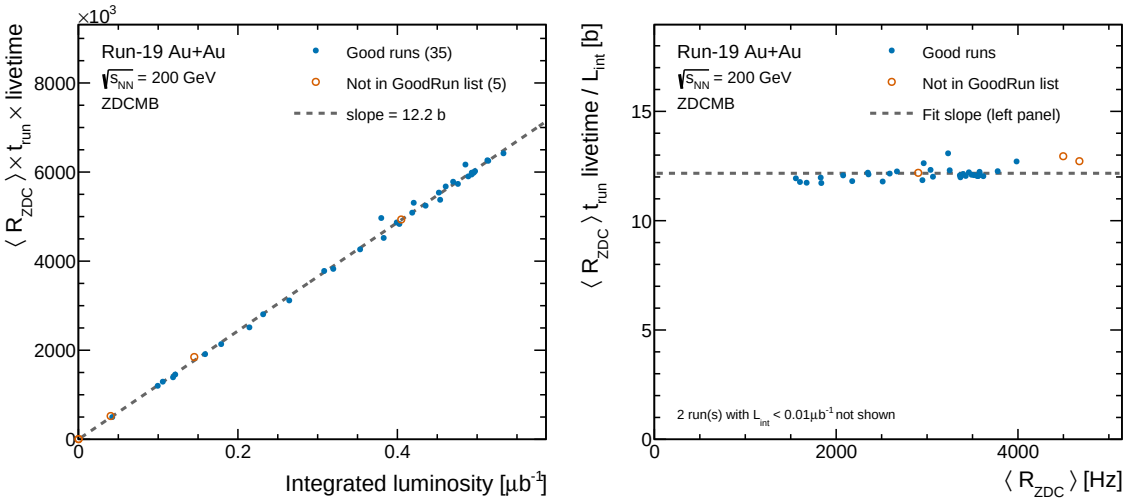


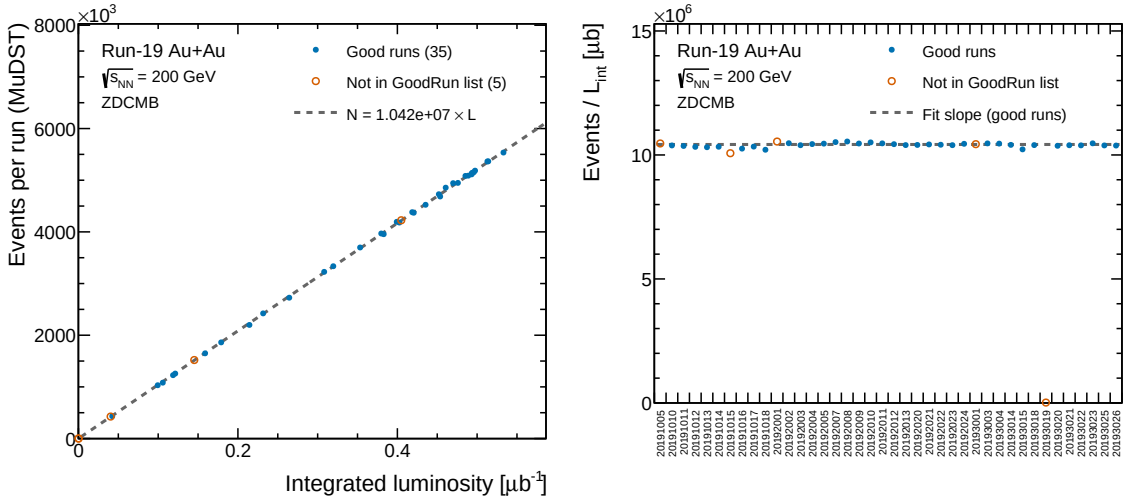
Figure 2: ZDC unattenuated ADC values for the Run-19 Au+Au zdcmb trigger.

2.1 Luminosity

The integrated luminosity for the zdcmb trigger is obtained from the STAR luminosity database for Run-19, where it is determined from the ZDC coincidence rate and the ZDC coincidence cross section. The corresponding net cross section, extracted as the slope of the live ZDC coincidences in each run, $\langle R_{\mathrm{ZDC}} \rangle \times t_{\mathrm{run}} \times$ livetime, versus the database luminosity, is 12.2 b (Figure 3); here t_{run} is the run duration. The number of recorded events versus the integrated luminosity for each run is shown in Figure 4 and exhibits a strong linear correlation, indicating that the luminosity determination is consistent for all analyzed runs. The integrated luminosity for the zdcmb trigger before bad-run removal is $L_{\mathrm{int}} = 13.2666\,\mu\mathrm{b}^{-1}$, with a prescale factor of 1.



62 Figure 3: Live ZDC coincidences per run versus integrated luminosity (left), and their ratio,
63 the implied ZDC cross section, versus the mean ZDC coincidence rate (right).



64 Figure 4: Recorded events per run versus integrated luminosity (left), and events per unit
65 luminosity for each run (right).

66 2.2 Bad Runs

67 Prior to applying event-level selections, runs with known detector-quality issues are
68 removed from the analysis. The standard STAR BES-II Run QA bad-run list for Run-19
69 Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV is used to exclude periods with degraded TPC
70 performance. Table 1 lists the run-by-run integrated luminosities for the bad runs.

Table 1: Integrated luminosity associated with the BES-II QA bad runs removed from the Run-19 Au+Au dataset.

Run number	Integrated luminosity [μb^{-1}]
20191005	0.145248
20191015	0.000152
20192001	0.040369
20193001	0.404860
20193019	0.000067
Total	0.590695

The nominal integrated trigger luminosity after bad-run removal for this analysis is

$$L_{\text{nominal}} = 12.6759\,\mu b^{-1} \tag{2}$$

3 Event Selection

UPCs are characterized by large impact parameters, which strongly suppress direct hadronic interactions between the two nuclei. In photon-induced vector-meson production, the exchanged photon is colorless and the interaction with the target proceeds through a color-singlet exchange. Consequently, no color connection is established between the two beam nuclei, suppressing particle production between the centrally produced vector meson and the outgoing nuclear remnants. This gives rise to large rapidity gaps on either side of the produced ϕ meson.

We exploit this topology to suppress hadronic Au+Au backgrounds. The rapidity-gap requirement is implemented through a veto on forward charged-particle activity in the Beam-Beam Counters (BBCs), while the exclusive nature of the final state is further selected using low central-event multiplicity requirements.

3.1 Vertex Selection

The reconstructed primary vertex is used to identify the collision location and to ensure that selected events originate within the nominal STAR interaction region. Since the geometrical acceptance and tracking performance of the TPC depend on the vertex position, particularly along the beam direction, events reconstructed far from the detector center may exhibit reduced or non-uniform tracking efficiency. Vertex-quality requirements are therefore applied to define a well-controlled collision region and to suppress poorly reconstructed or non-collision vertices before the remaining event-level selections are imposed.

For this analysis, we require the reconstructed event vertex to satisfy

$$|V_z| \leq 100\,\text{cm}, \qquad V_r \leq 5\,\text{cm}, \tag{3}$$

where $V_r = \sqrt{V_x^2 + V_y^2}$ is the transverse vertex position. The radial requirement restricts the vertex to the beam region. The integrated luminosity is determined using the ZDC-based luminosity measurement, whose longitudinal acceptance is not restricted in the same way as the TPC tracking acceptance. Because the effective fiducial region of the TPC is smaller

than the longitudinal interaction region sampled by the ZDC luminosity measurement, the V_z requirement introduces an additional fiducial acceptance on the analyzed event sample. The corresponding efficiency associated with this restricted vertex region, including the V_z -dependent vertex reconstruction and tracking efficiency, is therefore evaluated separately using embedding and incorporated into the analysis correction.

3.2 BBC Veto

The BBCs are forward scintillator detectors located on the east and west sides of STAR, approximately 3.7 m from the interaction point. The small (inner) BBC tiles are available in this dataset and cover the pseudorapidity range $3.3 < |\eta| < 5.0$. The BBCs are therefore sensitive to charged-particle production in the forward region. In this analysis, BBC activity is used as a veto to select the large rapidity gaps characteristic of UPC events and to suppress backgrounds from hadronic Au+Au collisions.

In past STAR UPC triggers, the BBC-veto online trigger was utilized for rapidity gap enforcement with the small BBC tiles. For each of the 18 small-tile PMTs on a given side, a BBC hit was defined by:

$$\mathrm{ADC}_{\mathrm{PMT}} > 20 \tag{4}$$

together with the timing requirement

$$100 \leq \mathrm{TDC}_{\mathrm{PMT}} \leq 2300. \tag{5}$$

The qualifying PMT ADC signals (those with a BBC hit) were summed separately in the East and West,

$$S_{\mathrm{East}} \equiv \sum_{e\in \text{ hits}} \mathrm{ADC}_e, \quad S_{\mathrm{West}} \equiv \sum_{w\in \text{ hits}} \mathrm{ADC}_w, \tag{6}$$

and the UPC-Main online veto threshold was 50 ADC channels on each side,

$$S_{\mathrm{East}} \leq 50 \quad \wedge \quad S_{\mathrm{West}} \leq 50. \tag{7}$$

The zdcmb trigger used in this analysis does not require this veto, but the BBC conditions are still evaluated at Level 0 and recorded with every event. The online BBC veto decision is obtained directly from the Level-0 trigger information stored in `StTriggerData`. For the Run-19 UPC zdcmb configuration, the East and West BBC conditions correspond to global DSM bits 17 and 18, respectively. Since `lastDSM(1)` contains global trigger bits 16–31, these correspond to bits 1 and 2 of the returned 16-bit word. The quantities are extracted as

$$\texttt{bbcEast} = (\texttt{lastDSM(1)} >> 1)\ \&\ 1, \qquad \texttt{bbcWest} = (\texttt{lastDSM(1)} >> 2)\ \&\ 1. \tag{8}$$

A value of one indicates that the corresponding BBC activity condition fired, i.e. that the small-tile BBC signal exceeded the online veto threshold on that side. An event therefore passes the BBC veto when neither the East nor the West bit is set. This analysis applies exactly this online requirement offline,

$$\texttt{bbcEast} = 0 \qquad \text{and} \qquad \texttt{bbcWest} = 0. \tag{9}$$

3.3 Event Multiplicity Selection

To select events consistent with the exclusive two-track final state of $\phi \to K^+K^-$, an event-multiplicity requirement is applied in addition to the BBC veto. The corresponding selections are:

- Each event must contain exactly two primary tracks, $N_{\mathrm{prim}} = 2$.
- The two primary tracks must be associated with the same reconstructed primary vertex.

These requirements select events with a single two-track primary vertex and suppress events containing additional charged-particle activity, as expected from hadronic interactions or pileup.

4 Track Selection and Particle Identification

After the event selection, each candidate event contains a single primary vertex with exactly two primary tracks. This section characterizes these tracks through their tracking quality, kinematics, and specific energy loss ($\mathrm{d}E/\mathrm{d}x$) in the TPC, and establishes the track-quality requirement and the kaon identification used by the candidate selections of Section 5. Since the final state is reconstructed from only two tracks in an otherwise empty detector, the track sample is already relatively clean. We therefore apply a minimal set of track-quality requirements, chosen to retain as many good tracks as possible while rejecting poorly reconstructed ones.

Particle identification is based on the TPC $\mathrm{d}E/\mathrm{d}x$. For a track i and a particle hypothesis $X \in \{e, \pi, K, p\}$, the normalized deviation of the measured energy loss from the expected one is

$$n\sigma_X^{(i)} = \frac{1}{\sigma_{\mathrm{d}E/\mathrm{d}x}} \ln \frac{(\mathrm{d}E/\mathrm{d}x)^{(i)}_{\mathrm{meas}}}{(\mathrm{d}E/\mathrm{d}x)^{(i)}_X}, \tag{10}$$

where $(\mathrm{d}E/\mathrm{d}x)^{(i)}_X$ is the Bichsel expectation for hypothesis X at the momentum of track i and $\sigma_{\mathrm{d}E/\mathrm{d}x}$ is the $\mathrm{d}E/\mathrm{d}x$ resolution. For the two tracks of a pair, the compatibility with the hypothesis that both tracks are of type X is quantified by

$$\chi^2_{XX} = \left(n\sigma_X^{(1)}\right)^2 + \left(n\sigma_X^{(2)}\right)^2, \tag{11}$$

which is evaluated for the kaon (χ^2_{KK}), pion ($\chi^2_{\pi\pi}$), electron (χ^2_{ee}), and proton (χ^2_{pp}) hypotheses.

The analysis tuple contains the pairs that pass the following preselection:

- zdcmb trigger (700001), bad runs removed;
- exactly two primary tracks associated with the pair's vertex;
- pair $\mathrm{d}E/\mathrm{d}x$ compatibility with the kaon–kaon hypothesis, $\chi^2_{KK} < 100$ (Equation 11), computed from the uncorrected $n\sigma_K$ (Section 4.2.1);
- track $p_T < 1\,\mathrm{GeV}/c$ and pair $p_T^{KK} < 1\,\mathrm{GeV}/c$;
- pair invariant mass, computed with the kaon mass hypothesis, $M_{KK} < 1.2\,\mathrm{GeV}/c^2$;
- no TOF requirement.

Unless stated otherwise, the distributions in this section also require the BBC veto, $N_{\mathrm{prim}} = 2$, and $|V_z| \leq 100\,\mathrm{cm}$ (Section 3). The two tracks form an unlike-sign (ULS) pair, the signal sample, or a like-sign (LS) pair, which is kept as a background reference.

4.1 Track Quality

4.1.1 Primary-Vertex Reconstruction

Primary vertices in this production (SL23e) are reconstructed with the Minuit vertex finder (`StMinuitVertexFinder`), which fits the vertex position from the distances of closest approach of TPC global tracks. Only global tracks passing `StMinuitVertexFinder::accept()` enter the fit:

```
bool StMinuitVertexFinder::accept(StTrack* track) const
{
  return (track &&
          track->flag() >= 0 &&
          track->fitTraits().numberOfFitPoints() >= mMinNumberOfFitPointsOnTrack
&&
          !track->topologyMap().trackFtpc() &&
          finite(track->length()) &&
          track->geometry()->helix().valid());
}
```

The cut values come from the `Calibrations/tracker/PrimaryVertexCuts` database table (Table 2).

Table 2: Track and vertex requirements of the Minuit vertex finder for the Run-19 Au+Au production.

Parameter	Value	Requirement
`MinNumberOfFitPointsOnTrack`	20	global-track fit points $N_{\mathrm{fit}}^{\mathrm{glob}} \geq 20$
`RImpactMax`	6 cm	transverse distance of the global helix to the beamline $< 6\,\mathrm{cm}$
`DcaZMax`	6 cm	track z at closest approach within 6 cm of the vertex seed
`MinTrack`	2	at least two accepted tracks per vertex
`ZMin`, `ZMax`	-200, 205 cm	range of the vertex search along z
`MinTrackPt`	0	not applied
`MinFracOfPossFitPointsOnTrack`	0	not applied

Since each selected vertex has exactly two tracks, both must pass $N_{\mathrm{fit}}^{\mathrm{glob}} \geq 20$. The primary refit adds the vertex as a fit point, giving the sharp edge at $N_{\mathrm{fit}} = 21$ (Figure 5, Figure 6); the small tail below it consists of tracks that lost hits in this refit. We therefore adopt

$$N_{\mathrm{hits}} \geq 20 \tag{12}$$

as the nominal track-quality selection, with $N_{\text{hits}} = N_{\text{fit}} - 1$ the number of TPC fit points. Its efficiency is evaluated with embedding.

The vertex finder applies **no minimum p_T requirement**: `MinTrackPt` is zero and is not read by the Minuit finder. The only implicit low-p_T limit is that strongly curling tracks may not reach 20 fit points; this is included in the embedding efficiency.

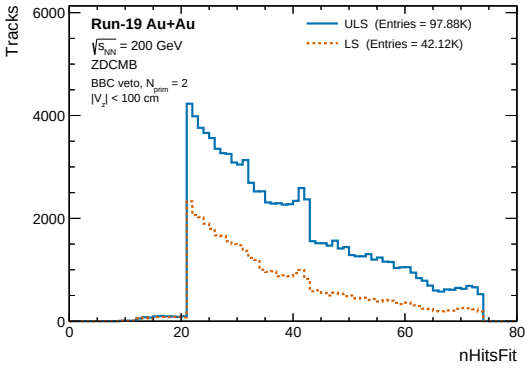


Figure 5: Number of fit points of primary tracks in ULS and LS pairs.

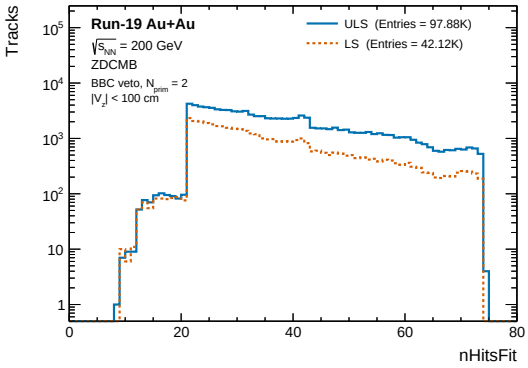


Figure 6: Same as Figure 5 on a logarithmic scale.

4.1.2 Inner TPC Upgrade and the Hit Requirement

The full iTPC (Section 2) increased the inner-sector pad rows from 13 to 40 and the maximum number of points per track from 45 to 72. The iTPC is fully operational in this sample: N_{fit} extends well beyond 45 (Figure 5), N_{poss} peaks at the full inner-sector and full-TPC lengths (Figure 7), and nearly all tracks have iTPC hits (Figure 8).

This makes $N_{\text{hits}} \geq 20$ a mild requirement here, although it is tighter than the 15 hits common in STAR UPC analyses. Many tracks, mostly at large $|\eta|$, leave through the endcap without reaching the outer sectors; with 40 inner rows they still collect 20 hits, which produces the edge at 20 in Figure 8, whereas with 13 inner rows they could not.

The fit-point threshold is not new: `MinNumberOfFitPointsOnTrack` is 20 also in 2012, 2014, and 2018. What changed is `MinTrack`, which was 5 in those years and is 2 here, so the Minuit finder can now reconstruct exclusive two-track events.

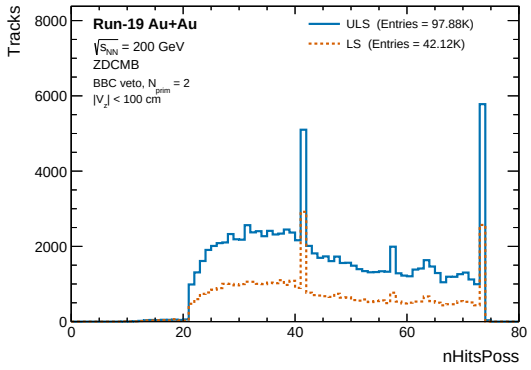


Figure 7: Number of possible points of primary tracks in ULS and LS pairs.

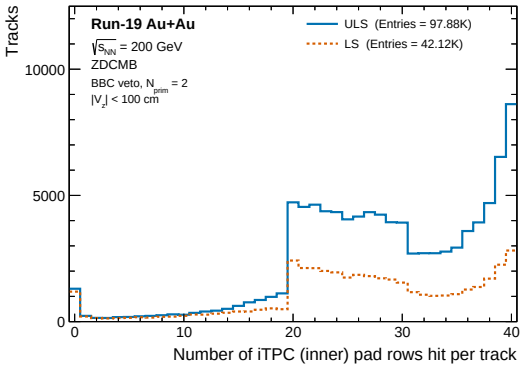


Figure 8: Number of iTPC pad rows hit per track.

4.2 Particle Identification

Figure 9 shows $\mathrm{d}E/\mathrm{d}x$ versus momentum for all tracks (ULS and LS). The curves give the expected $\mathrm{d}E/\mathrm{d}x$ for each hypothesis, taken as the mean $\mathrm{d}E/\mathrm{d}x$ of tracks with $|n\sigma_X| < 0.3$, i.e. the reference used in Equation 10. At low momentum the kaon band is well separated from the pion band, while the two merge above $p \approx 1\,\mathrm{GeV}/c$.

The sample is nevertheless dominated by pions. The preselection $\chi^2_{KK} < 100$ only limits each track to $|n\sigma_K| < 10$ (Figure 10), which at these momenta still accepts most of the pion band. This is also visible in the one-dimensional distributions: $n\sigma_\pi$ peaks sharply at zero, while $n\sigma_K$ is dominated by a broad pion contribution at negative values (Figure 11, Figure 12). A dedicated kaon selection is therefore required (Section 5.2.3, Section 5.3.2). Figure 10 also shows that the kaon band is centered at positive $n\sigma_K$ rather than at zero; this shift is measured with TOF-identified kaons in Section 4.2.1.

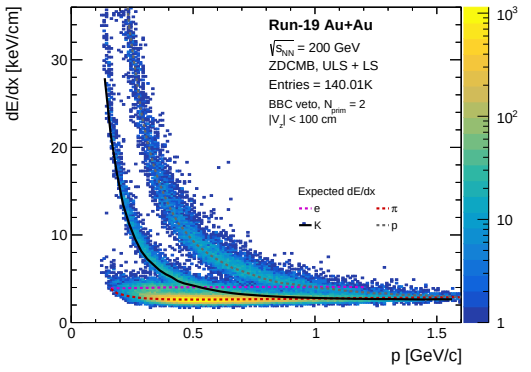


Figure 9: $\mathrm{d}E/\mathrm{d}x$ versus momentum for all tracks, with the expected $\mathrm{d}E/\mathrm{d}x$ for each hypothesis.

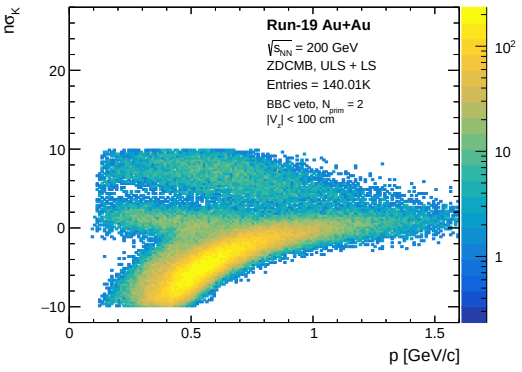


Figure 10: $n\sigma_K$ versus momentum for the same tracks.

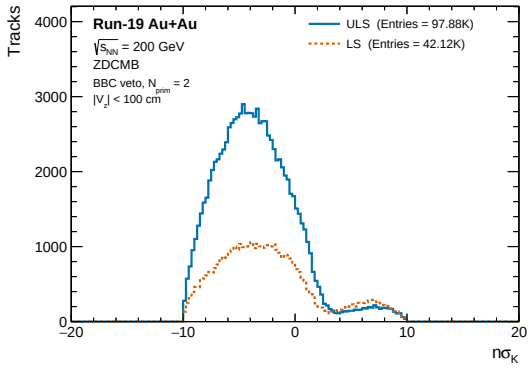


Figure 11: $n\sigma_K$ of tracks in ULS and LS pairs.

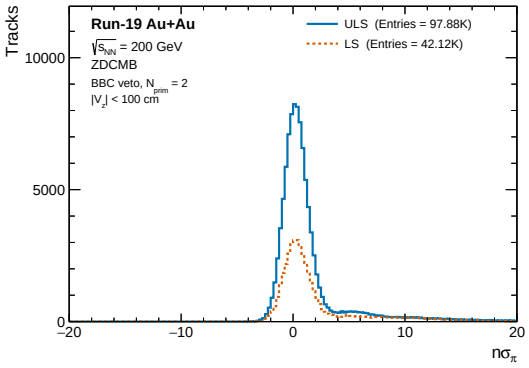


Figure 12: $n\sigma_\pi$ of the same tracks.

4.2.1 $n\sigma_K$ Calibration with TOF-Identified Kaons

The $n\sigma_K$ shift is measured with kaons identified independently of $\mathrm{d}E/\,\mathrm{d}x$ by the TOF. The $\phi \to K^+K^-$ UPC sample itself yields few TOF-identified kaons: the decay kaons are soft and mostly at large $|\eta|$, outside the TOF acceptance, and the sample is small. We therefore use kaons from peripheral hadronic Au+Au collisions in the same Run-19 data (zdcmb trigger), referred to below as the peripheral sample, which cover the full momentum range with high statistics. The TOF-identified kaons are selected as follows:

- centrality 70–80% from `StRefMultCorr` (`getCentralityBin9() == 0`), $|V_z| < 50\,\mathrm{cm}$, $V_r < 2\,\mathrm{cm}$, bad runs removed;
- primary tracks of the highest-ranked vertex with $N_{\mathrm{hits}} \geq 20$ and $N_{\mathrm{hits}}/N_{\mathrm{poss}} > 0.52$;
- TOF-matched with $\beta < 1$ and $0.22 < m^2 < 0.27\,\mathrm{GeV}^2/c^4$, where $m^2 = p^2(1/\beta^2 - 1)$ (the kaon tuple extends to $0.28\,\mathrm{GeV}^2/c^4$);
- $|n\sigma_K| < 5$.

The TOF mass spectrum (Figure 13) shows well-separated pion, kaon, and proton peaks, and the m^2 window selects the kaon peak (Figure 14). The selected tracks form a single band in $n\sigma_K$ over the full momentum range (Figure 15), with only small pion and proton contamination at low momentum, confirming that the sample consists of genuine kaons.

The tracks are divided into momentum bins of $0.1\,\mathrm{GeV}/c$ between 0.1 and $2.5\,\mathrm{GeV}/c$, and the $n\sigma_K$ distribution in each bin is fitted with a Gaussian in $[-5, 5]$. The fitted mean $\langle n\sigma_K \rangle(p)$ (Figure 16) is positive at all momenta and largest at low momentum, consistent with the shift seen in Figure 10. We correct for it with

$$n\sigma_K^{\mathrm{corr}} = n\sigma_K - \langle n\sigma_K \rangle(p). \tag{13}$$

All kaon requirements from here on use $n\sigma_K^{\mathrm{corr}}$, and χ^2_{KK} denotes the value computed from it.

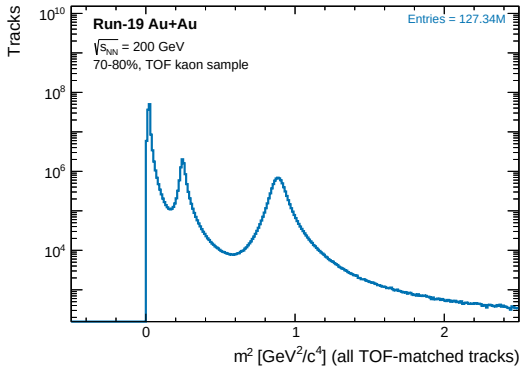


Figure 13: TOF m^2 of all TOF-matched tracks in the peripheral sample.

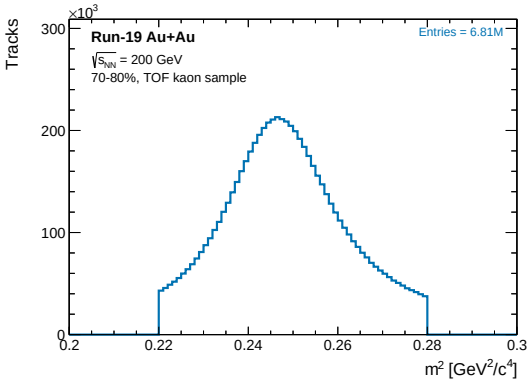


Figure 14: m^2 of the kaon candidates in the tuple ($0.22 < m^2 < 0.28\,\mathrm{GeV}^2/c^4$).

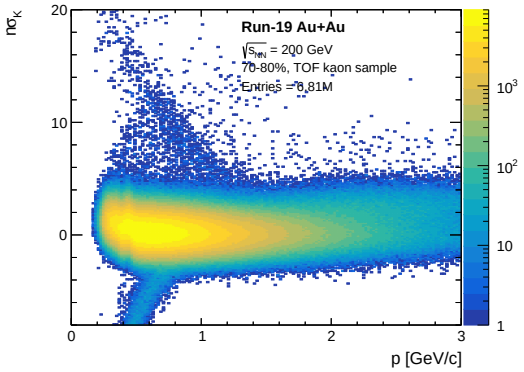


Figure 15: $n\sigma_K$ versus momentum for the same kaon candidates, before the $|n\sigma_K| < 5$ requirement.

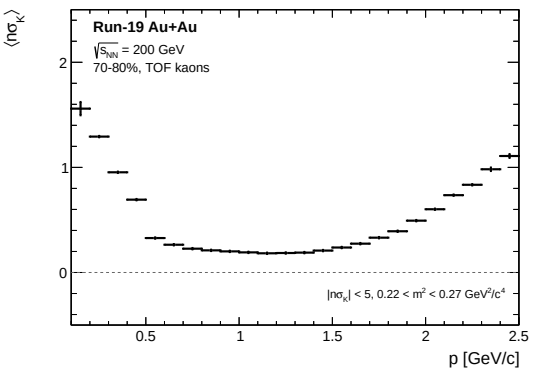


Figure 16: Mean $n\sigma_K$ of TOF-identified kaons versus momentum, from Gaussian fits.

The Gaussian fit in each momentum bin up to $p = 1.6\,\mathrm{GeV}/c$ is shown in Figure 17. The correction is applied in this range, which contains essentially all tracks of the ϕ sample; at higher momentum the kaon and pion bands merge and the fits become unreliable.

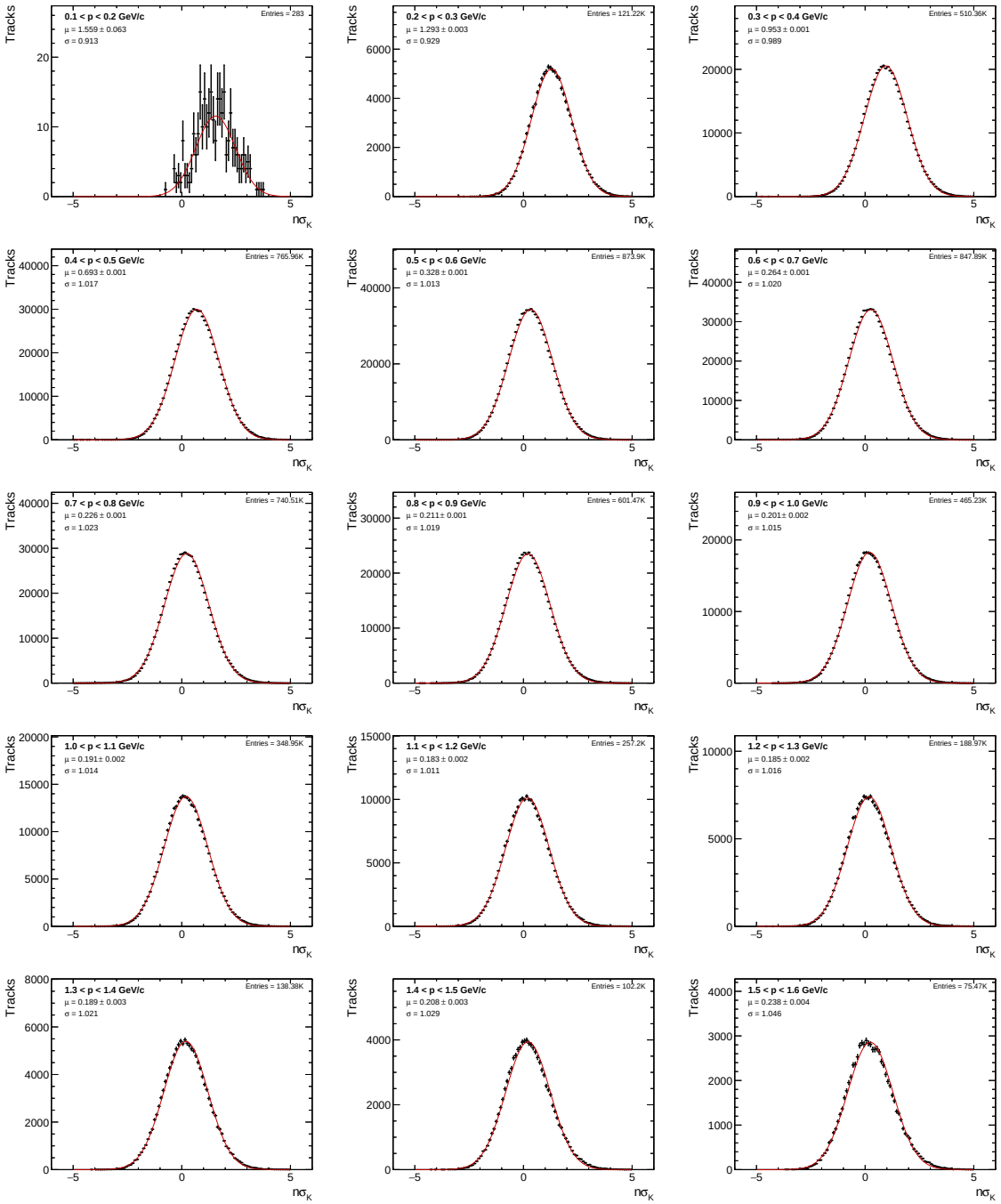


Figure 17: $n\sigma_K$ of TOF-identified kaons in each momentum bin with the Gaussian fit (red) used for Figure 16.

4.2.2 Cross-Check with Kaons from $\phi \to K^+K^-$

The $n\sigma_K$ shift is cross-checked in situ with kaons from ϕ decays in the UPC sample, selected as follows:

- BBC veto, $N_{\text{prim}} = 2$, $|V_z| \leq 100\,\text{cm}$, $V_r \leq 5\,\text{cm}$, pair rapidity $|y_{KK}| < 0.5$, and $|n\sigma_K| \leq 5$ for both tracks;
- tag and probe: a track (probe) is used when the other track (tag) satisfies the pion rejection $|n\sigma_\pi| \geq 3$, so that no pion rejection is applied to the track whose $n\sigma_K$ is measured.

The non-ϕ background in the mass window $1.00 < M_{KK} < 1.04\,\text{GeV}/c^2$ is described by probes from the sideband $1.06 < M_{KK} < 1.10\,\text{GeV}/c^2$. The sideband is scaled to the background in the window obtained from a fit of the probe mass spectrum with a Voigtian (PDG width) plus a linear background (Figure 18). In each momentum bin, the $n\sigma_K$ distribution of the window probes is fitted with a binned Poisson likelihood using a Gaussian for the kaons plus the scaled sideband distribution as a fixed background template.

The resulting $\langle n\sigma_K \rangle(p)$ agrees with the TOF-identified kaons within the statistical uncertainties (Figure 19). Since the TOF sample is far more precise and covers the full momentum range, the TOF result (Section 4.2.1) is used as the nominal correction.

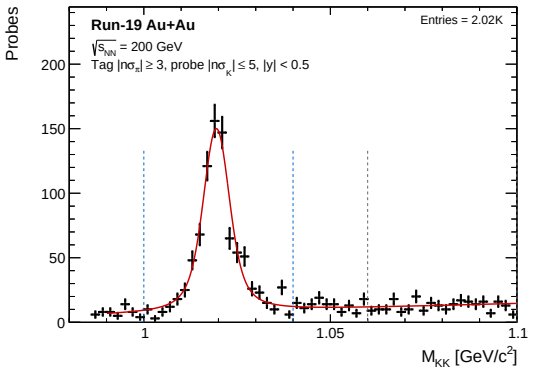


Figure 18: M_{KK} of tag-and-probe pairs with the fit; dashed lines mark the mass window and the sideband.

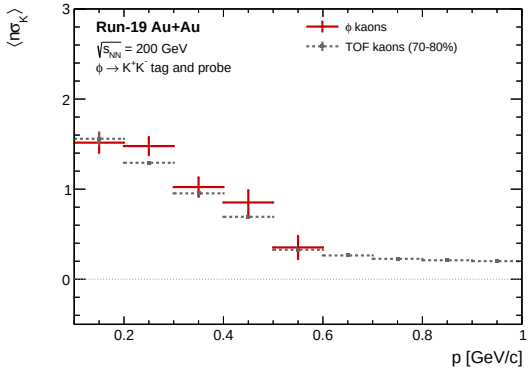


Figure 19: Mean $n\sigma_K$ of kaons from ϕ decays compared with the TOF-identified kaons.

5 Candidate Selection and Raw Yields

This section combines the event selection (Section 3) with the track requirements of Section 4 to select coherent and incoherent ϕ candidates. In both cases the raw yield is the number of ULS pairs in the mass window $1.00 < M_{KK} < 1.04\,\text{GeV}/c^2$, before background subtraction, acceptance and efficiency corrections, and luminosity normalization. The cutflow tables start from the preselected ULS pairs of Section 4 and count pairs over the full mass range, $M_{KK} < 1.2\,\text{GeV}/c^2$, and in the mass window. An $N-1$ distribution shows one variable with all other nominal requirements applied; no mass-window requirement is applied to these distributions. In the figures, dashed lines mark the cut values of the plotted variable, or the mass window in mass spectra.

5.1 Coherent and Incoherent Regions

In coherent production the photon couples to the nucleus as a whole, so the pair p_T is limited by the nuclear size, $p_T^{KK} \lesssim \hbar c/R_{\mathrm{Au}}$; incoherent production on individual nucleons extends to much larger p_T^{KK}. We define the two regions by a cut on the pair p_T alone:

$$\text{coherent: } p_T^{KK} \leq 0.1\,\mathrm{GeV}/c, \qquad \text{incoherent: } 0.3 < p_T^{KK} < 0.9\,\mathrm{GeV}/c. \tag{14}$$

The upper limit of the incoherent region follows from the electron-rejection study of Section 5.3.3.

Figure 20 shows the pair p_T before any kaon identification. The ULS spectrum has a pronounced coherent peak below $0.1\,\mathrm{GeV}/c$ that is absent in the LS pairs, and the two regions are well separated by the p_T cut alone. This separation depends only on the nuclear form factor and not on the produced vector meson, so it holds even though the sample is still dominated by pions at this stage.

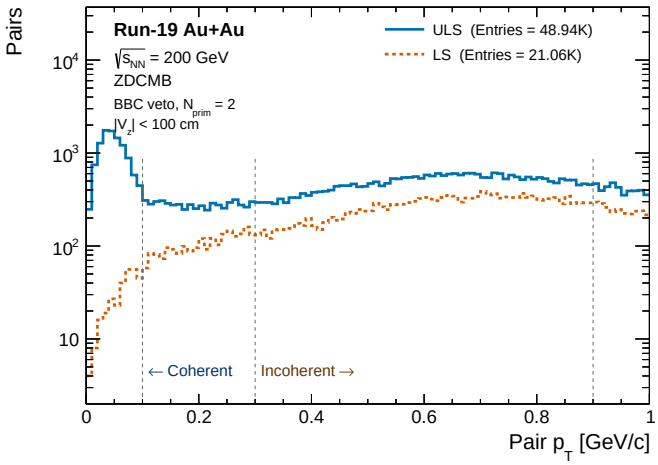


Figure 20: Pair p_T of ULS and LS pairs, with the coherent and incoherent regions of Equation 14.

5.2 Coherent Candidates

5.2.1 Selection

Table 3 lists the nominal coherent selection, and Table 4 gives the cumulative ULS pair counts. The pair, kaon-identification, and track-momentum requirements are motivated in Section 5.2.2, Section 5.2.3, and Section 5.2.4. Figure 21 and Figure 22 show the $N-1$ distributions of the vertex coordinates.

Table 3: Nominal coherent selection.

Level	Requirement
Event	BBC veto; $N_{\mathrm{prim}} = 2$; $\lvert V_z \rvert \leq 100\,\mathrm{cm}$; $V_r \leq 5\,\mathrm{cm}$
Pair	ULS; $p_T^{KK} \leq 0.1\,\mathrm{GeV}/c$; $\lvert y_{KK} \rvert < 0.5$
Track quality	Both tracks: $N_{\mathrm{hits}} \geq 20$
PID	$\chi^2_{KK} \leq 15$
Track momentum	Both tracks: $p_T \leq 0.20\,\mathrm{GeV}/c$
Mass window	$1.00 < M_{KK} < 1.04\,\mathrm{GeV}/c^2$

Table 4: Cumulative ULS pair counts for the coherent selection.

Cumulative selection	All masses	Mass window
Preselected ULS pairs	1922337	329903
BBC veto	313439	55367
$N_{\mathrm{prim}} = 2$	74804	11507
$\lvert V_z \rvert \leq 100\,\mathrm{cm}$; $V_r \leq 5\,\mathrm{cm}$	46561	6489
Both tracks: $N_{\mathrm{hits}} \geq 20$	45873	6332
$p_T^{KK} \leq 0.1\,\mathrm{GeV}/c$	10209	97
$\lvert y_{KK} \rvert < 0.5$	724	68
$\chi^2_{KK} \leq 15$	143	61
Both tracks: $p_T \leq 0.20\,\mathrm{GeV}/c$	69	61

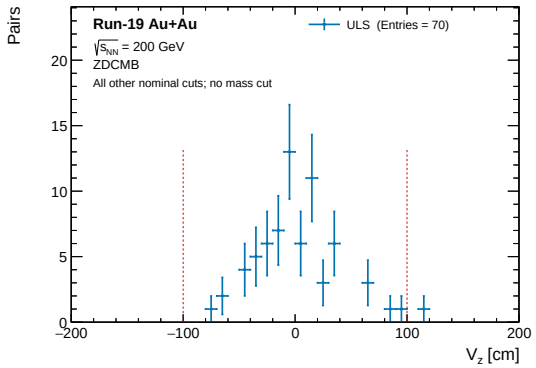


Figure 21: $N-1$ distribution of V_z for coherent candidates.

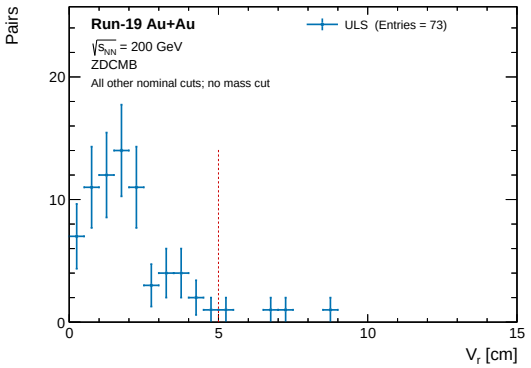


Figure 22: $N-1$ distribution of V_r for coherent candidates.

5.2.2 Pair Transverse Momentum and Rapidity

The requirement $p_T^{KK} \leq 0.1\,\mathrm{GeV}/c$ selects the coherent region of Equation 14 (Figure 23).

The pair-rapidity window is chosen from the mass spectra in four signed-rapidity intervals (Figure 25 to Figure 28), with all other nominal requirements applied except the

track-momentum requirement. Mass-window pairs occur only in the three inner intervals, $-0.5 \leq y_{KK} \leq 0.5$; the interval $0.5 < y_{KK} \leq 1$ contains none. The full rapidity distribution (Figure 24) shows that no mass-window pair lies at $|y_{KK}| > 0.5$ on either side. We therefore use $|y_{KK}| < 0.5$.

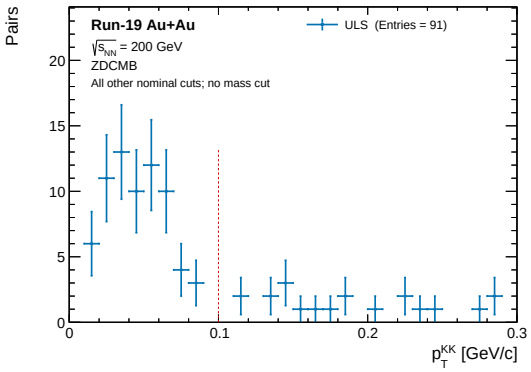


Figure 23: $N-1$ distribution of p_T^{KK} for coherent candidates.

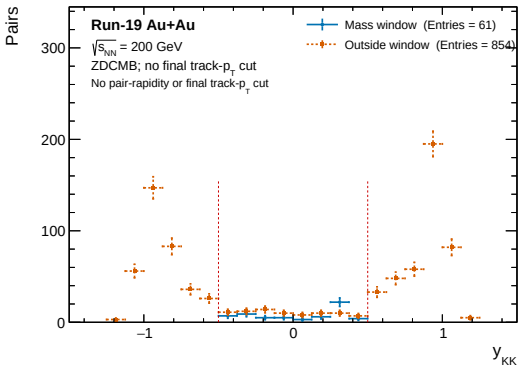


Figure 24: Pair rapidity inside and outside the mass window, before the track-momentum requirement.

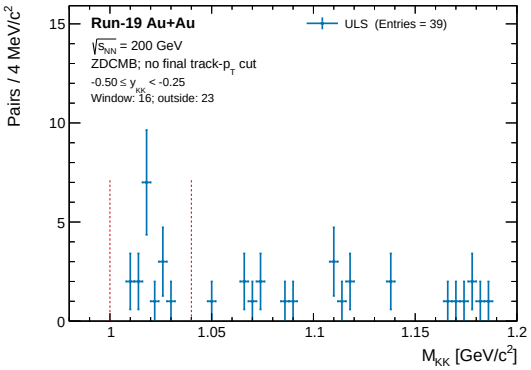


Figure 25: ULS mass spectrum for $-0.5 \leq y_{KK} < -0.25$, without the rapidity and track-momentum requirements.

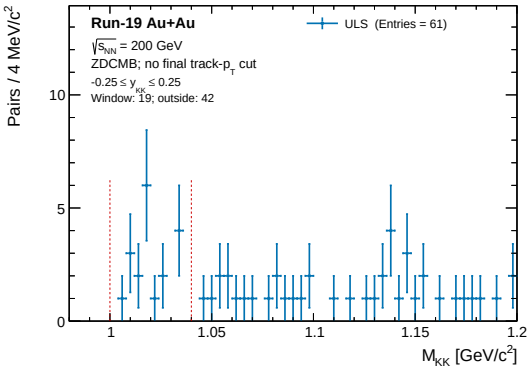


Figure 26: Same as Figure 25 for $|y_{KK}| \leq 0.25$.

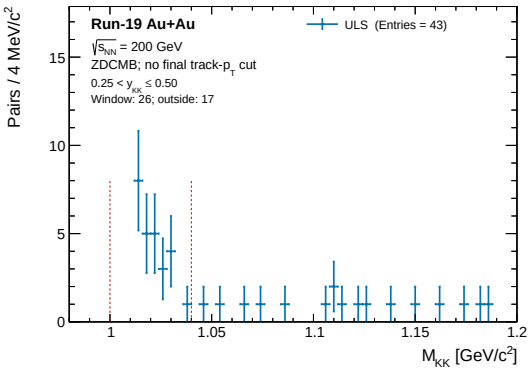


Figure 27: Same as Figure 25 for $0.25 < y_{KK} \leq 0.5$.

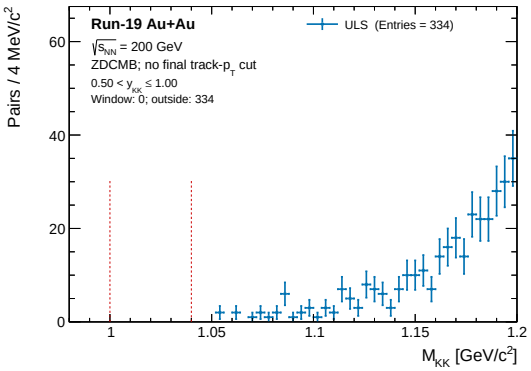


Figure 28: Same as Figure 25 for $0.5 < y_{KK} \leq 1$.

5.2.3 Kaon Identification

In the coherent region the decay kaons have momenta below about $0.3\,\mathrm{GeV}/c$ (Figure 77), where the kaon and pion $\mathrm{d}E/\,\mathrm{d}x$ bands are well separated (Figure 9). A loose requirement, $\chi^2_{KK} \leq 15$, is therefore applied. Figure 29 shows its $N-1$ distribution.

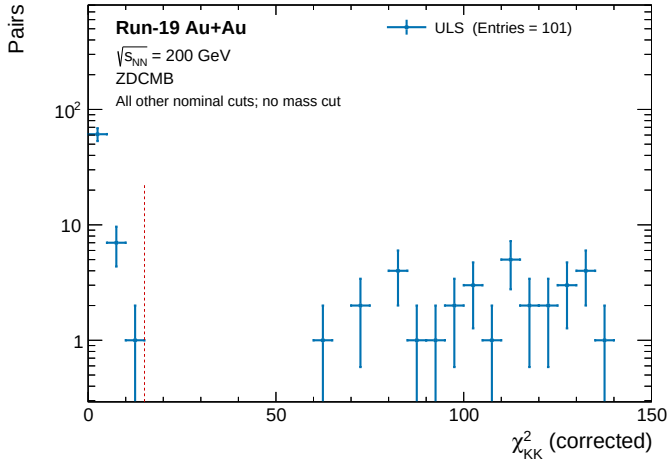


Figure 29: $N-1$ distribution of χ^2_{KK} for coherent candidates.

5.2.4 Track Transverse Momentum

The pair p_T^{KK} is the magnitude of the vector sum of the two track transverse momenta, so a low pair p_T does not imply that each track has low p_T. Figure 30 shows the larger of the two track momenta, and Figure 31 the mass spectrum before and after requiring $p_T \leq 0.20\,\mathrm{GeV}/c$ for both tracks. For nearly back-to-back tracks, the track-momentum requirement is correlated with M_{KK} and suppresses the higher-mass population, while it keeps all mass-window pairs. We adopt $p_T \leq 0.20\,\mathrm{GeV}/c$ for both tracks.

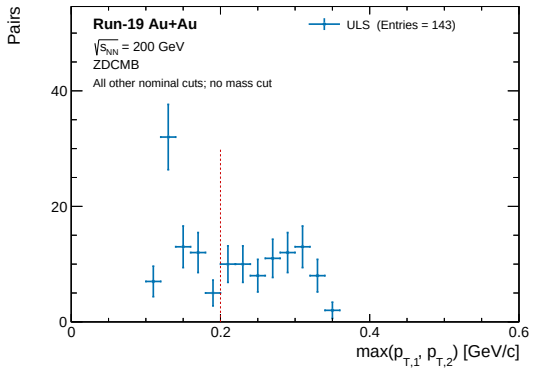


Figure 30: $N-1$ distribution of the larger track p_T for coherent candidates.

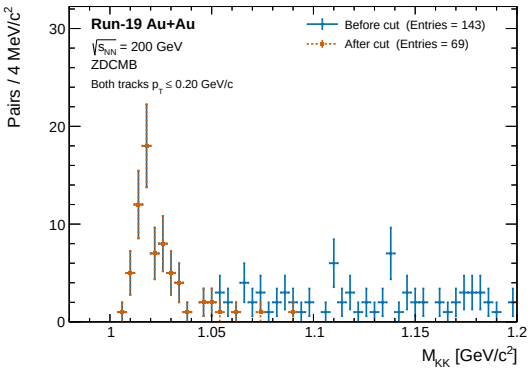


Figure 31: Coherent mass spectrum before and after the track-momentum requirement.

5.2.5 Mass Spectrum and Raw Yield

Figure 32 shows the mass spectrum after the nominal coherent selection. The number of ULS pairs in the mass window is the coherent raw yield,

$$N_{\text{raw}}^{\text{coh}} = 61. \tag{15}$$

Figure 33, Figure 34, and Figure 35 show the pair distributions of the mass-window pairs; their track distributions are collected in Appendix A.1.

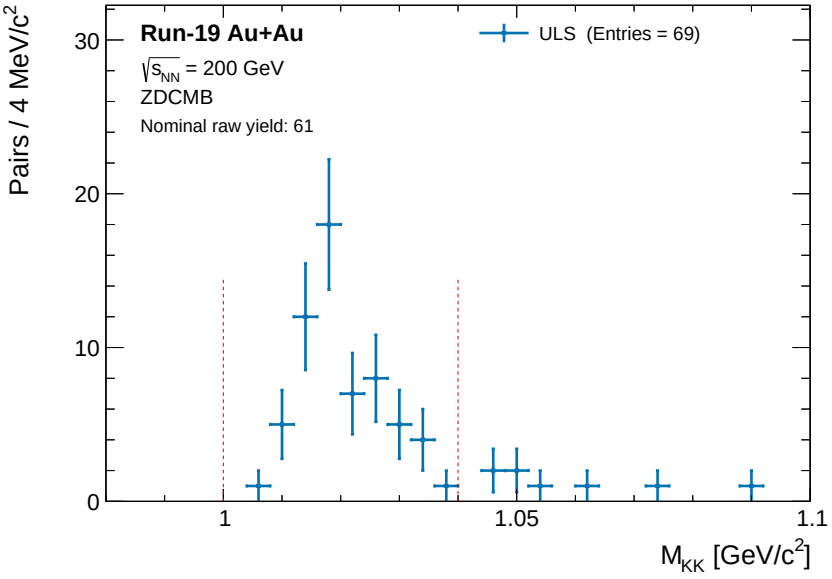


Figure 32: Invariant-mass spectrum after the nominal coherent selection.

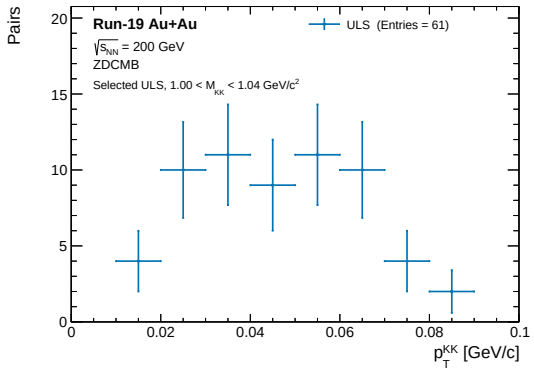


Figure 33: p_T^{KK} of the coherent mass-window pairs.

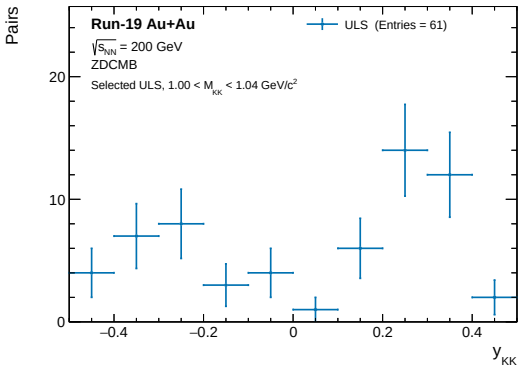


Figure 34: y_{KK} of the coherent mass-window pairs.

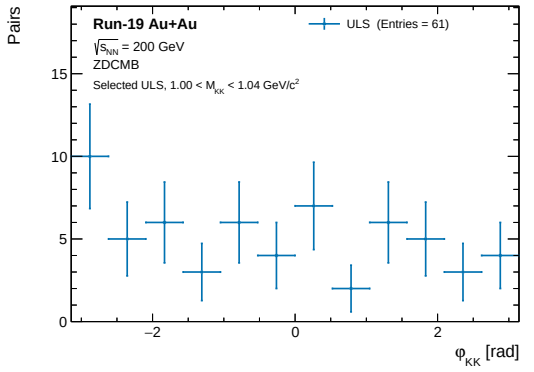


Figure 35: Pair azimuth of the coherent mass-window pairs.

5.3 Incoherent Candidates

5.3.1 Selection

Table 5 lists the nominal incoherent selection, and Table 6 gives the cumulative ULS pair counts. The event, vertex, and track-quality requirements and the rapidity interval are those of the coherent selection, and there is no track-momentum requirement beyond the preselection. The higher parent p_T shifts the daughter momenta upward, where the TPC response bands approach each other, so the PID is chosen in a separate study (Section 5.3.2, Section 5.3.3). Figure 36 and Figure 37 show the $N-1$ distributions of the pair requirements.

Table 5: Nominal incoherent selection.

Level	Requirement
Event	BBC veto; $N_{\mathrm{prim}} = 2$; $\lvert V_z \rvert \leq 100\,\mathrm{cm}$; $V_r \leq 5\,\mathrm{cm}$
Pair	ULS; $0.3 < p_T^{KK} < 0.9\,\mathrm{GeV}/c$; $\lvert y_{KK} \rvert < 0.5$
Track quality	Both tracks: $N_{\mathrm{hits}} \geq 20$
PID	$\chi^2_{KK} \leq 10$; $\chi^2_{\pi\pi} \geq 15$; no electron or proton rejection
Track momentum	No requirement beyond the preselection
Mass window	$1.00 < M_{KK} < 1.04\,\mathrm{GeV}/c^2$

Table 6: Cumulative ULS pair counts for the incoherent selection.

Cumulative selection	All masses	Mass window
Preselected ULS pairs	1922337	329903
BBC veto	313439	55367
$N_{\mathrm{prim}} = 2$	74804	11507
$\lvert V_z \rvert \leq 100\,\mathrm{cm}$; $V_r \leq 5\,\mathrm{cm}$	46561	6489
Both tracks: $N_{\mathrm{hits}} \geq 20$	45873	6332
$0.3 < p_T^{KK} < 0.9\,\mathrm{GeV}/c$	26557	5390
$\lvert y_{KK} \rvert < 0.5$	5604	1156
$\chi^2_{KK} \leq 10$	540	298
$\chi^2_{\pi\pi} \geq 15$	534	298

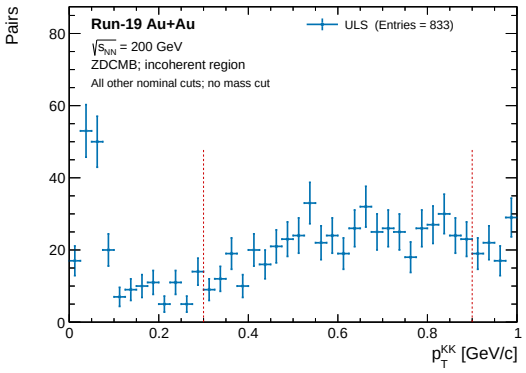


Figure 36: $N-1$ distribution of p_T^{KK} for incoherent candidates.

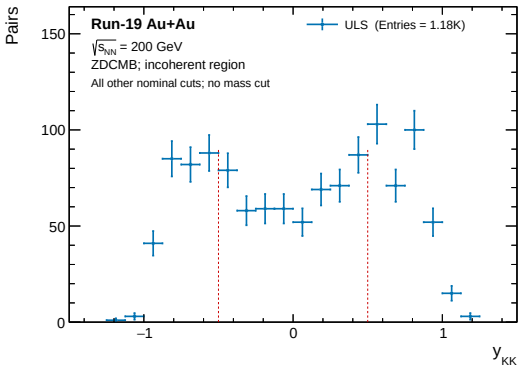


Figure 37: $N-1$ distribution of y_{KK} for incoherent candidates.

5.3.2 Kaon Identification

The PID requirements are chosen from the ULS mass-window pairs that pass all nominal requirements except the PID. Their single-track $n\sigma$ and pair χ^2 distributions are shown in Figure 38 to Figure 45, before and after the nominal PID; the error bars of the single-track

distributions account for the correlation of the two tracks of a pair. The corrected kaon response contains a concentration near zero, together with a broad population away from the kaon expectation; the pion, electron, and proton hypotheses show how these candidates populate the other response bands.

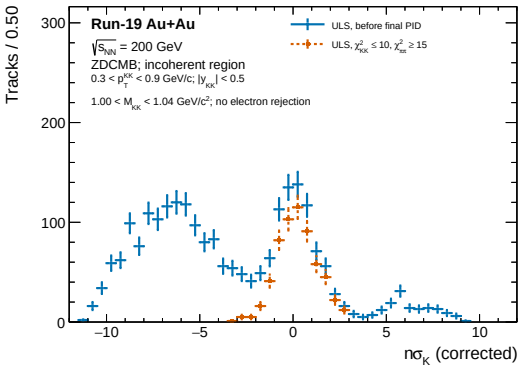


Figure 38: $n\sigma_K^{\mathrm{corr}}$ of tracks in incoherent mass-window pairs, before and after the nominal PID.

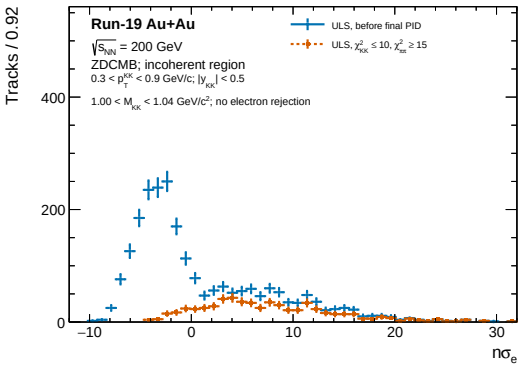


Figure 39: Same as Figure 38 for $n\sigma_e$.

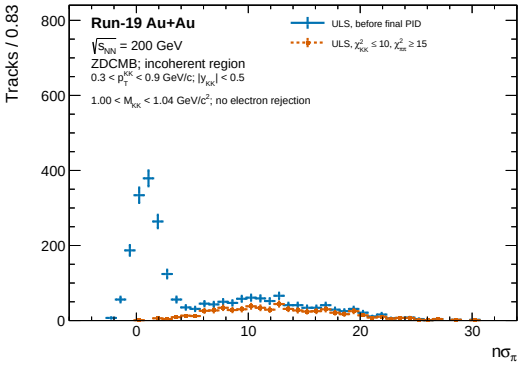


Figure 40: Same as Figure 38 for $n\sigma_\pi$.

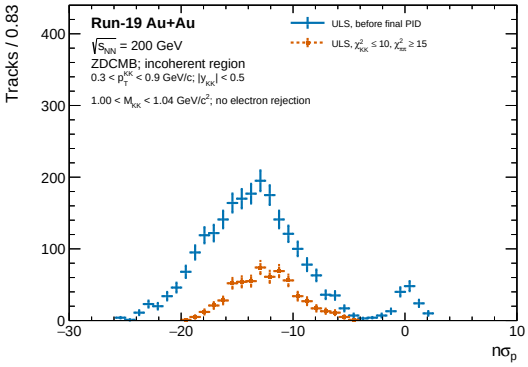


Figure 41: Same as Figure 38 for $n\sigma_p$.

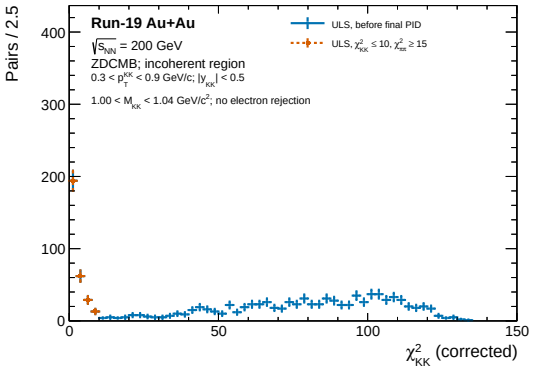


Figure 42: χ^2_{KK} of incoherent mass-window pairs, before and after the nominal PID.

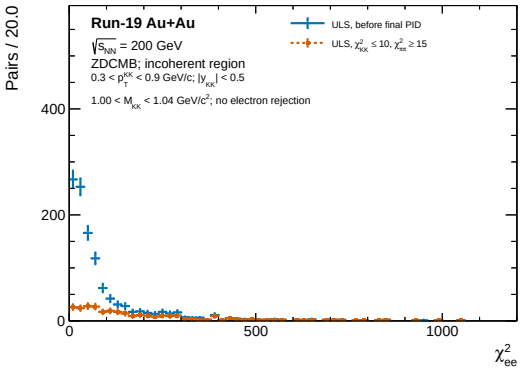


Figure 43: Same as Figure 42 for χ^2_{ee}.

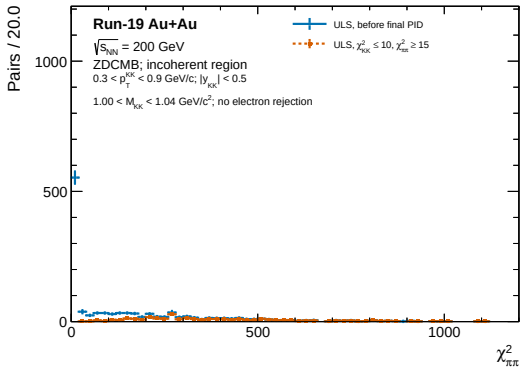


Figure 44: Same as Figure 42 for $\chi^2_{\pi\pi}$.

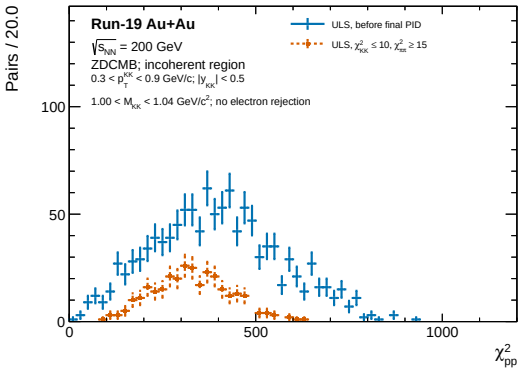


Figure 45: Same as Figure 42 for χ^2_{pp}.

Figure 46 shows the corrected kaon response of track 1 against that of track 2. The central population lies near the origin, separated from the broad population at negative $n\sigma_K^{\mathrm{corr}}$. The requirement

$$\chi^2_{KK} = \left(n\sigma^{\mathrm{corr}}_{K,1}\right)^2 + \left(n\sigma^{\mathrm{corr}}_{K,2}\right)^2 \leq 10 \tag{16}$$

selects a circle of radius $\sqrt{10}$ around the kaon expectation, which retains the central population without extending far into the surrounding tails. Table 7 compares nearby boundaries: tightening to 8 removes 11 mass-window pairs, while loosening to 15 adds 9 mass-window pairs and 70 pairs outside the window. We therefore use $\chi^2_{KK} \leq 10$.

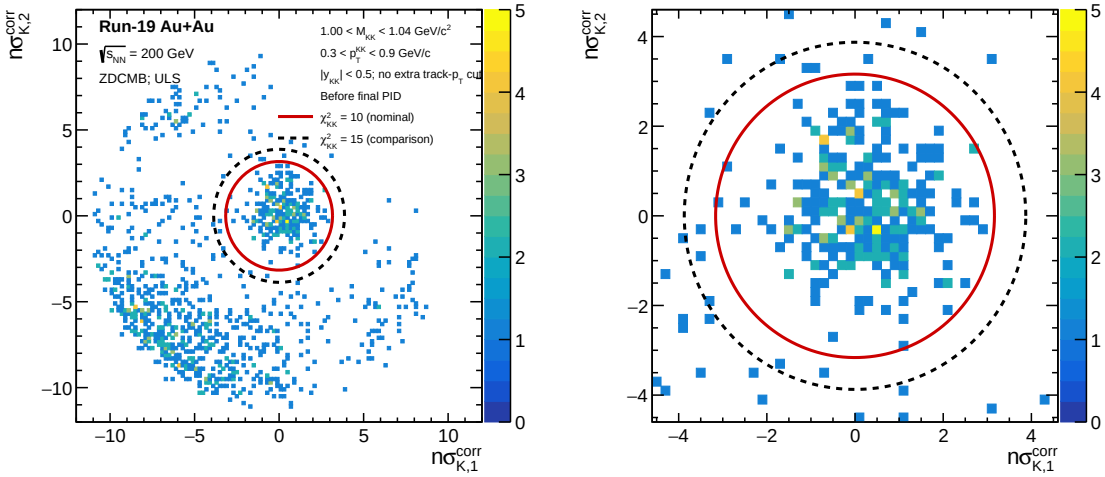


Figure 46: $n\sigma^{\mathrm{corr}}_{K,1}$ versus $n\sigma^{\mathrm{corr}}_{K,2}$ for incoherent mass-window pairs before PID, with circles at $\chi^2_{KK} = 10$ (solid) and 15 (dashed).

Table 7: ULS pair counts for three kaon requirements, without pion or electron rejection.

Kaon requirement	All masses	Mass window	Outside window
$\chi^2_{KK} \leq 8$	505	287	218
$\chi^2_{KK} \leq 10$	540	298	242
$\chi^2_{KK} \leq 15$	619	307	312

All mass-window pairs already satisfy $\chi^2_{\pi\pi} \geq 15$; this mild pion requirement is kept for background control and removes 6 pairs above the mass window.

5.3.3 Electron Rejection

At the larger daughter momenta of the incoherent sample, a kaon-compatible TPC response can also overlap the electron hypothesis. Figure 47 shows χ^2_{KK} versus χ^2_{ee} in the mass window; some candidates that pass the kaon requirement would fail one of the electron requirements

$$\chi^2_{ee} \geq 10, \qquad \chi^2_{ee} \geq 2\chi^2_{KK}. \tag{17}$$

The mass spectrum retains a clear enhancement near the ϕ mass without electron rejection (Figure 48). Applying both electron requirements removes 13 of the 298 mass-window pairs, $(4.4 \pm 1.2)\%$, including pairs in the ϕ peak (Figure 49). Since the ϕ is close to the KK threshold, electron pairs reconstructed under the kaon mass hypothesis can also populate this region. We therefore test with TOF-identified kaons whether this loss is expected from the kaon $\mathrm{d}E/\mathrm{d}x$ response itself.

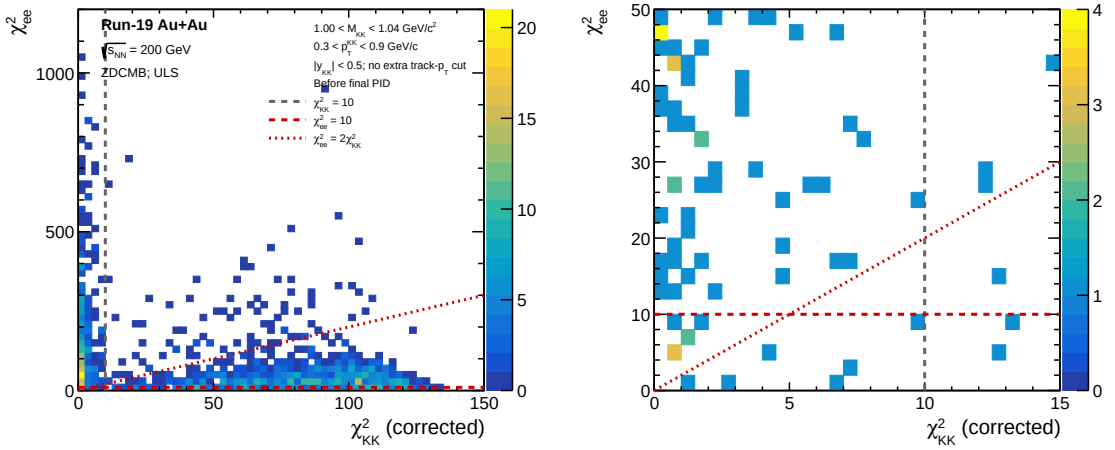


Figure 47: χ^2_{KK} versus χ^2_{ee} in the incoherent mass window before PID, with the kaon requirement (vertical line) and the electron requirements of Equation 17.

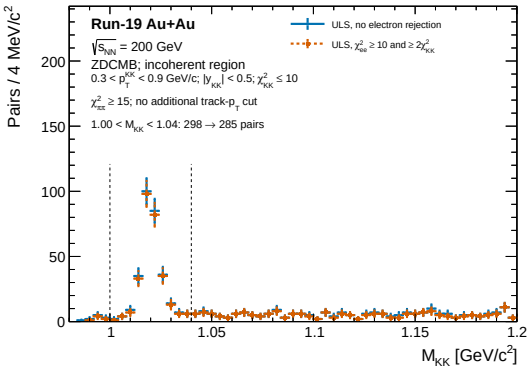


Figure 48: Incoherent mass spectrum with and without the electron requirements.

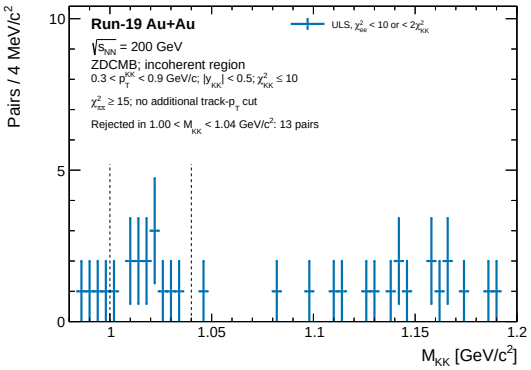


Figure 49: Mass spectrum of the pairs removed by the electron requirements.

5.3.3.1 Prediction from TOF-Identified Kaons

The prediction asks how often true kaons with the measured daughter momenta would fail the electron requirements. It uses kaon candidates from the peripheral sample (Section 4.2.1) with $0.22 < m^2 < 0.28\,\mathrm{GeV}^2/c^4$, $N_{\mathrm{hits}} \geq 20$, and $0.1 \leq p < 1.0\,\mathrm{GeV}/c$, without any TPC $n\sigma$ requirement: 5.2 million tracks, binned by charge and in $0.01\,\mathrm{GeV}/c$ of momentum. Their $n\sigma_K$ is corrected in the same way as for the UPC candidates.

For each mass-window pair, one donor track is drawn for each daughter, matched in charge and momentum bin. A donor supplies its measured $(n\sigma_K^{\mathrm{corr}}, n\sigma_e, n\sigma_\pi)$ together, which preserves their correlations and the measured response tails; the two daughters are drawn independently. For each pair, 4000 donor pairs are drawn. Among the draws that pass the preselection $\chi^2_{KK} < 100$ and the nominal kaon and pion requirements, the fraction that fails either electron requirement gives the rejection probability of pair i and the expected number of rejected pairs,

$$P_i^{\mathrm{loss}} = \frac{N_i^{\mathrm{rejected}}}{N_i^{\mathrm{accepted}}}, \qquad N_{\mathrm{predicted}}^{\mathrm{rejected}} = \sum_i P_i^{\mathrm{loss}}. \tag{18}$$

Pairs with fewer than 30 kaon donors in either daughter's momentum bin, or fewer than 30 accepted draws, are excluded from both the data and the prediction.

The pool covers 268 of the 298 mass-window pairs. In this common sample the electron requirements reject 13 pairs, against 14.6 predicted: a loss of $(4.9 \pm 1.3)\%$ in data and $(5.4 \pm 1.2)\%$ predicted. The prediction uncertainty includes the spread of the measured momenta and the finite number of draws. The dependence on pair p_T also agrees: nearly all of the rejection occurs at $0.7 < p_T^{KK} < 0.9\,\mathrm{GeV}/c$ (Table 8, Figure 50).

Table 8: Observed and predicted electron-requirement losses, in pairs, for the mass-window pairs covered by the TOF-identified kaons.

p_T^{KK} [GeV/c]	Covered pairs	Observed loss	Predicted loss
0.3–0.5	51 / 73	0	0.00
0.5–0.7	118 / 126	1	1.27
0.7–0.9	99 / 99	12	13.30
Full interval	268 / 298	13	14.57

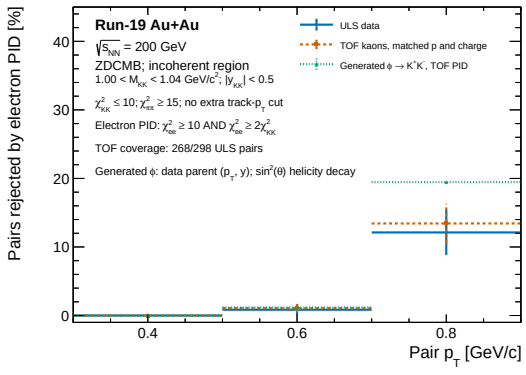


Figure 50: Electron-requirement loss versus p_T^{KK} for data, the TOF-kaon prediction, and generated ϕ decays.

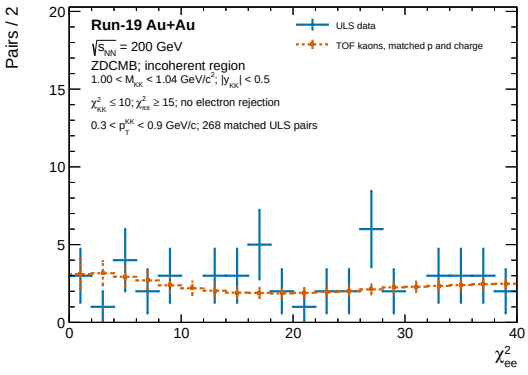


Figure 51: Measured and predicted χ^2_{ee} distributions of the covered mass-window pairs.

As a kinematic cross-check, 2 million $\phi \to K^+K^-$ decays are generated with the parent (p_T, y) drawn jointly from the 298 mass-window pairs before electron rejection (including any background in the window), a Breit-Wigner mass restricted to the mass window, a uniform parent azimuth, and a $\sin^2\theta$ decay distribution in the helicity frame. The daughters receive donor responses at their generated momenta and charges. Unlike the prediction above, which uses the measured daughter momenta, this check generates new daughter momenta from the parent kinematics; its loss, evaluated in the part of the generated sample covered by the TOF-identified kaons, is shown with the data and the prediction in Figure 50.

The transfer assumes that charge- and momentum-matched TOF-identified kaons represent the TPC response of the decay kaons. The agreement with the observed loss, including its concentration at high pair p_T, supports kaon-response overlap as the explanation for the removed candidates in the ϕ peak. We therefore apply no electron rejection.

The loss grows with p_T^{KK} as the daughters reach the momentum range where the kaon and electron responses overlap: in $0.7 < p_T^{KK} < 0.9\,\mathrm{GeV}/c$ the prediction from the measured daughters gives $(13.4 \pm 2.9)\%$ and the generated decays 19.5%. This rising loss sets the upper limit $p_T^{KK} < 0.9\,\mathrm{GeV}/c$ of the incoherent region (Equation 14).

5.3.4 Mass Spectrum and Raw Yield

Figure 52 shows the mass spectrum after the nominal incoherent selection. The number of ULS pairs in the mass window is the incoherent raw yield,

$$N_{\mathrm{raw}}^{\mathrm{incoh}} = 298. \tag{19}$$

Figure 53, Figure 54, and Figure 55 show the pair distributions of the mass-window pairs; their track distributions are collected in Appendix A.2.

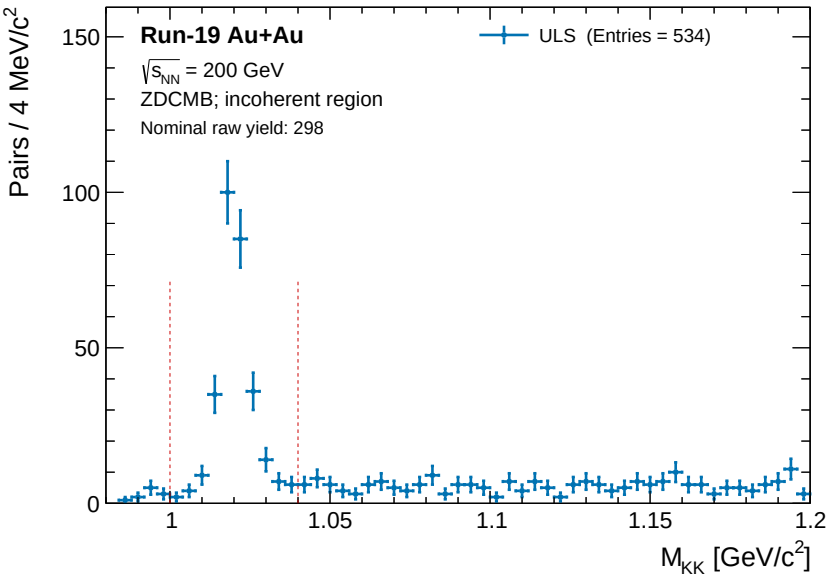


Figure 52: Invariant-mass spectrum after the nominal incoherent selection.

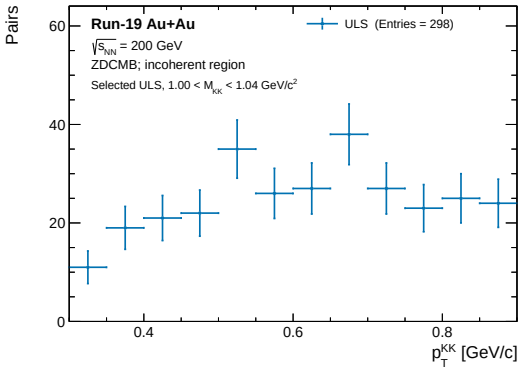


Figure 53: p_T^{KK} of the incoherent mass-window pairs.

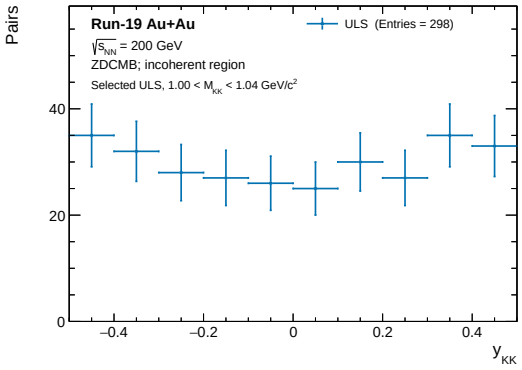


Figure 54: y_{KK} of the incoherent mass-window pairs.

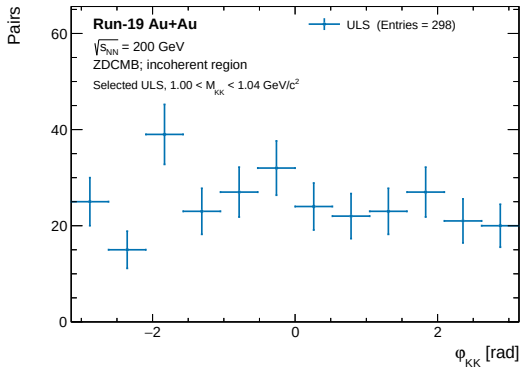


Figure 55: Pair azimuth of the incoherent mass-window pairs.

6 Efficiencies

6.1 Vertex Selection Efficiency

The vertex selection efficiency is the fraction of well-reconstructed vertices from the luminous region that lie inside the fiducial region $|V_z| \leq 100$ cm. It is measured with two samples: all UPC-like events (Section 6.1.1), whose V_z distribution has a broad background plateau, and a clean sample of coherent di-pion events (Section 6.1.2). Both give an efficiency consistent with unity, and $\varepsilon_{V_z} = 1.00$ is used as the nominal value (Section 6.1.3).

Both samples start from the event selection of Section 3 without the vertex requirements, referred to below as UPC-like events:

- zdcmb trigger, bad runs removed.
- `bbcEast` $= 0$ and `bbcWest` $= 0$.
- Exactly two primary tracks matched to the same vertex.

6.1.1 Using All UPC-like Events

For all UPC-like events, there is no kinematic or particle-identification restriction on the reconstructed vertices, so secondary interactions are not rejected. Figure 56 and Figure 57 show signatures of interactions at $V_r \approx 3.8$ cm, which correspond to the beryllium beam pipe radius.

A second, sharp structure appears at $V_r = 10$ cm and extends up to $V_r = 10\sqrt{2} \approx 14.1$ cm. In Figure 57 these vertices lie on the edges of the square $|V_x| = 10$ cm or $|V_y| = 10$ cm, which is consistent with vertices pushed to the boundary of the transverse region allowed in the vertex reconstruction rather than interactions at a physical radius. These vertices carry no information on the luminous region and are not considered further.

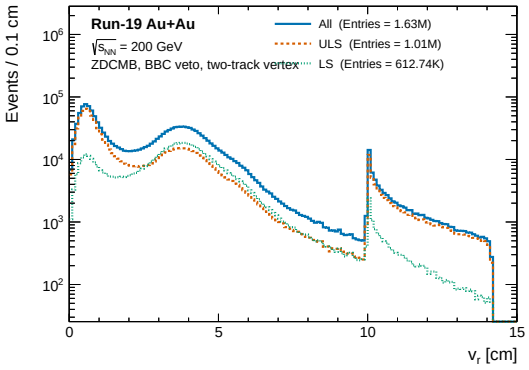


Figure 56: V_r distribution for all UPC-like events.

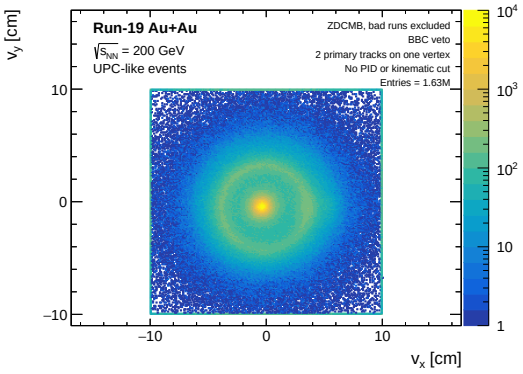


Figure 57: V_x versus V_y distribution for all UPC-like events.

To suppress secondary interactions for the efficiency study only, the following additional vertex selection criteria are implemented:

- $V_r < 3$ cm.
- Unlike-sign pairs only.

Figure 58 shows that the additional vertex quality cuts remove this background. Figure 59 and Figure 60 show the V_z distribution before and after the vertex quality cuts.

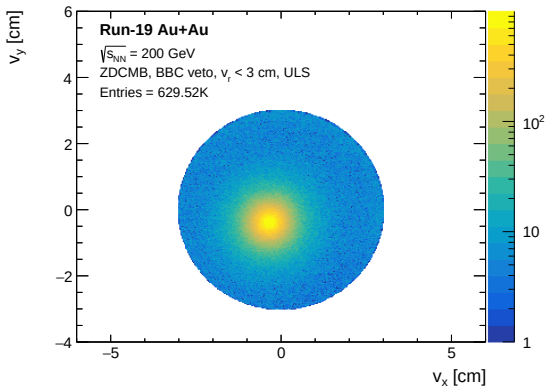


Figure 58: V_x versus V_y distribution for unlike-sign pairs in UPC-like events with $V_r < 3$ cm.

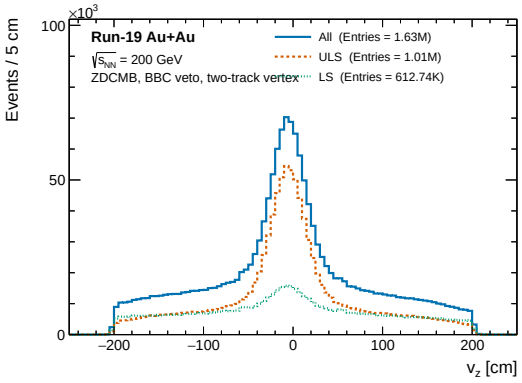


Figure 59: V_z distribution for all UPC-like events.

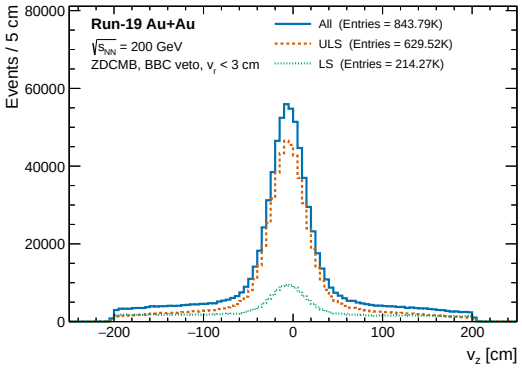


Figure 60: V_z distribution for UPC-like events with $V_r < 3$ cm.

The V_z distribution after the vertex quality cuts is fitted with the sum of two Gaussians,

$$f(V_z) = A_1 \exp\left(-\frac{(V_z-\mu_1)^2}{2\sigma_1^2}\right) + A_2 \exp\left(-\frac{(V_z-\mu_2)^2}{2\sigma_2^2}\right), \tag{20}$$

in 5 cm bins over $|V_z| < 190$ cm, which stays inside the ± 200 cm edge of the vertex reconstruction. The narrow Gaussian (σ_1) describes the luminous region, while the wide Gaussian (σ_2) absorbs the broad plateau seen in Figure 60. The fraction of a Gaussian inside the fiducial region is

$$F(\mu,\sigma) = \frac{1}{2}\left[\mathrm{erf}\left(\frac{100-\mu}{\sqrt{2}\sigma}\right) - \mathrm{erf}\left(\frac{-100-\mu}{\sqrt{2}\sigma}\right)\right], \tag{21}$$

from which three estimates of the fraction inside $|V_z| \le 100$ cm are formed:

- $f_{\mathrm{narrow}} = F(\mu_1, \sigma_1)$, from the narrow Gaussian only;
- $f_{\mathrm{both}} = (1-w_2)F(\mu_1,\sigma_1) + w_2 F(\mu_2,\sigma_2)$, from both Gaussians, where $w_2 = A_2\sigma_2/(A_1\sigma_1 + A_2\sigma_2)$ is the area fraction of the wide Gaussian;
- f_{count}, the fraction of entries with $|V_z| \le 100$ cm, counted directly from the histogram.

The fits to the unlike-sign pairs (Figure 61) and, as a cross-check, to the unlike-sign minus like-sign (ULS − LS) distribution (Figure 62) are summarized in Table 9. In the fit figures, dashed vertical lines mark $|V_z| = 100$ cm.

Table 9: Double-Gaussian fit results for UPC-like events with $V_r < 3$ cm. All fractions refer to $|V_z| \le 100$ cm.

Sample	σ_1 [cm]	σ_2 [cm]	w_2	f_{narrow}	f_{both}	f_{count}
ULS	19.9	124.3	0.372	1.0000	0.843	0.879
ULS − LS	19.0	64.1	0.242	1.0000	0.971	0.975

The narrow Gaussian is stable between the two samples ($\sigma_1 \approx 19$–20 cm, $\mu_1 \approx -5.4$ cm) and lies entirely inside the fiducial region, $f_{\mathrm{narrow}} = 1.0000$. The wide Gaussian, in contrast, changes substantially when the like-sign pairs are subtracted (σ_2 from 124 to 64 cm, w_2

from 0.37 to 0.24), showing that it is driven by the residual background plateau rather than by the luminous region. The lower values of f_{both} and f_{count} therefore reflect background that extends beyond $|V_z| = 100$ cm and are not used as the efficiency. The vertex selection efficiency from this method is taken from the narrow Gaussian, $\varepsilon_{V_z} \approx 1.000$. Both fits have large χ^2/ndf (964/70 and 340/70), as two Gaussians do not fully describe the peak-plus-plateau shape; this does not affect f_{narrow}, which is set by $\sigma_1 \approx 20$ cm relative to the 100 cm cut.

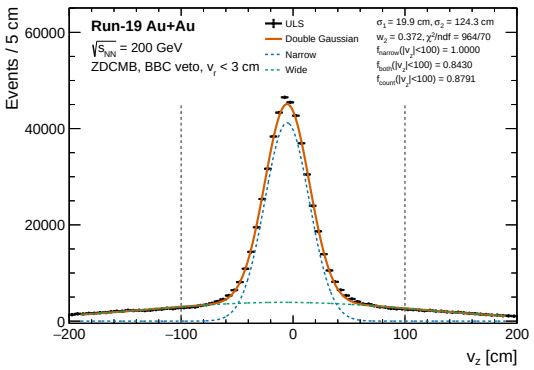


Figure 61: Double-Gaussian fit to the V_z distribution of ULS pairs in UPC-like events with $V_r < 3$ cm.

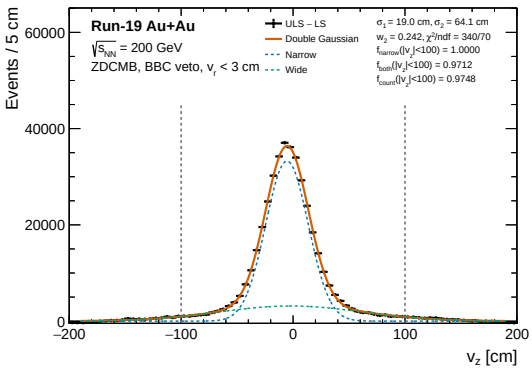


Figure 62: Double-Gaussian fit to the ULS − LS V_z distribution in UPC-like events with $V_r < 3$ cm.

6.1.2 Using Di-pion Events

To avoid the background plateau of the UPC-like events, the luminous region is also measured with a clean sample of coherently photoproduced $\pi^+\pi^-$ pairs. In addition to the UPC-like event definition, the following selection is applied:

- `TofMult` ≤ 6 and at least one track matched to the TOF.
- Both tracks satisfy $|n\sigma_\pi| < 3$.
- Pair transverse momentum $p_T^{\pi\pi} < 0.1$ GeV/c (coherent region).
- $0.60 < M_{\pi\pi} < 0.95$ GeV/c^2 around the ρ^0 pole, for the vertex distributions only.

The vertex coordinates are those of the vertex the two pions are matched to.

6.1.2.1 Di-pion Mass Spectrum

The mass spectrum in this section is not corrected for acceptance or efficiency and is not used to extract a cross section. It only serves to check that the selected di-pion sample is not background dominated and that its shape agrees with the published coherent ρ^0 result.

Figure 63 compares the unlike-sign and like-sign invariant-mass spectra of the selected pairs. The like-sign pairs are about two orders of magnitude below the unlike-sign pairs over the whole mass range. The unlike-sign minus like-sign spectrum is fitted over $0.6 < M_{\pi\pi} < 1.2$ GeV/c^2 with the modified Söding parametrization used by STAR for coherent ρ^0 photoproduction (Phys. Rev. C 96 (2017) 054904),

$$\frac{dN}{dM_{\pi\pi}} \propto \left| A_\rho \frac{\sqrt{M_{\pi\pi} M_\rho \Gamma_\rho}}{M_{\pi\pi}^2 - M_\rho^2 + i M_\rho \Gamma_\rho} + B_{\pi\pi} + C_\omega e^{i\varphi_\omega} \frac{\sqrt{M_{\pi\pi} M_\omega \Gamma_{\omega\to\pi\pi}}}{M_{\pi\pi}^2 - M_\omega^2 + i M_\omega \Gamma_\omega} \right|^2 + f_p, \quad (22)$$

with mass-dependent widths as in that reference and a linear remnant background f_p. The spectrum uses 2.5 MeV/c^2 bins in $0.72 < M_{\pi\pi} < 0.84$ GeV/c^2, where the ρ^0–ω interference is located, and 10 MeV/c^2 bins elsewhere. The fit and its components are shown in Figure 64 and the parameters are compared with the published values in Table 10.

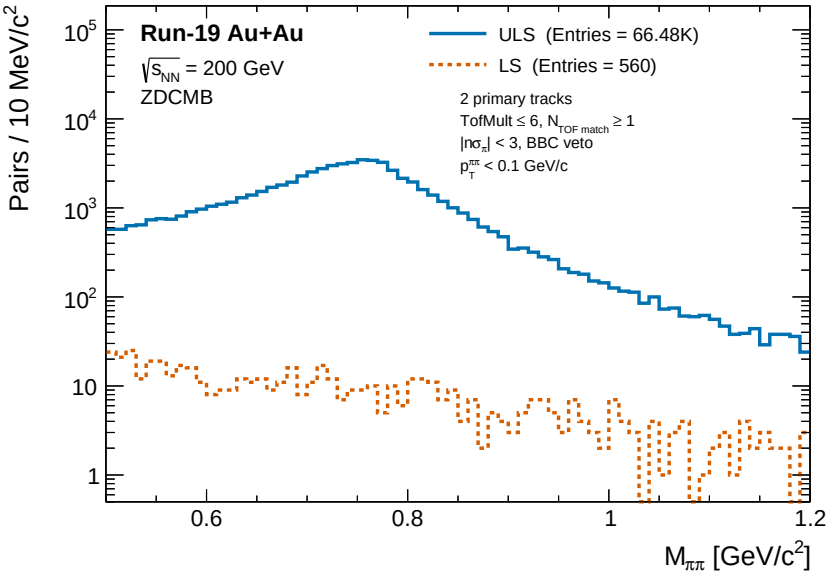
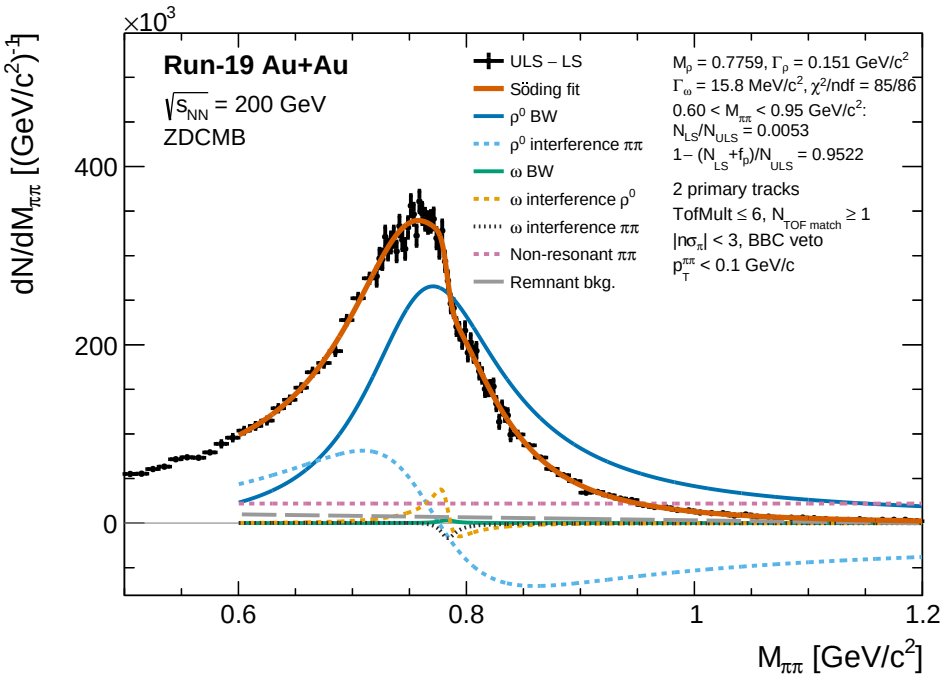


Figure 63: Invariant-mass spectra of unlike-sign and like-sign pion pairs passing the di-pion selection, without vertex or mass-window requirements.



610 Figure 64: Modified Söding fit to the ULS − LS $\pi^+\pi^-$ spectrum and its components.

611 Table 10: Modified Söding fit to the coherent unlike-sign minus like-sign $\pi^+\pi^-$ spectrum,
612 compared with the published fit. The published ratios are computed from the quoted
613 amplitudes.

614 Parameter	This analysis	STAR, PRC 96, 054904
615 M_ρ [GeV/c^2]	0.7759	0.7762 ± 0.0006
616 Γ_ρ [GeV/c^2]	0.151	0.156 ± 0.001
617 $B_{\pi\pi}/A_\rho$ [$(\mathrm{GeV}/c^2)^{-1/2}$]	−0.742	−0.787
618 C_ω/A_ρ	0.295	0.358
619 M_ω [GeV/c^2]	0.7829	0.7824 ± 0.0008
620 Γ_ω [MeV/c^2]	15.8	17 ± 2
621 φ_ω [rad]	1.33	1.46 ± 0.11
622 χ^2/ndf	85.5/86	–

623 The ρ^0 mass and width and the ω parameters are close to the published values. As the
624 spectrum is uncorrected and no systematic uncertainties are evaluated, this comparison
625 is qualitative: it shows that the lineshape is reproduced. As in the published fit, no mass
626 resolution is included in the model, so the fitted Γ_ω absorbs the detector resolution and is
627 about twice the natural width. The ratio $B_{\pi\pi}/A_\rho = -0.742$ is also close to the published
628 −0.787, showing that the non-resonant $\pi^+\pi^-$ component is described by $B_{\pi\pi}$ rather
629 than absorbed into the remnant background f_p. Inside the mass window $0.60 < M_{\pi\pi} <$

0.95 GeV/c^2 there are 56788 unlike-sign and 301 like-sign pairs ($N_{\mathrm{LS}}/N_{\mathrm{ULS}} = 0.0053$), and the remnant background f_p amounts to 2415 pairs, giving a $\pi^+\pi^-$ purity of $1 - (N_{\mathrm{LS}} + f_p)/N_{\mathrm{ULS}} = 0.952$. The selected sample is therefore dominated by coherent $\pi^+\pi^-$ production.

6.1.2.2 Vertex Distributions and Efficiency

Figure 65 and Figure 66 show the transverse position of the di-pion vertices. Unlike the UPC-like events, there is no ring at $V_r \approx 3.8$ cm: the vertices are concentrated around the beam spot, with only a small population at the $V_r = 10$ cm reconstruction boundary. The V_z distribution (Figure 67) shows no plateau, and the like-sign contribution is negligible.

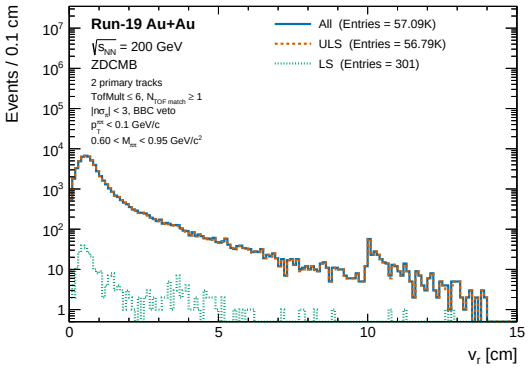


Figure 65: V_r distribution of coherent $\pi^+\pi^-$ pairs.

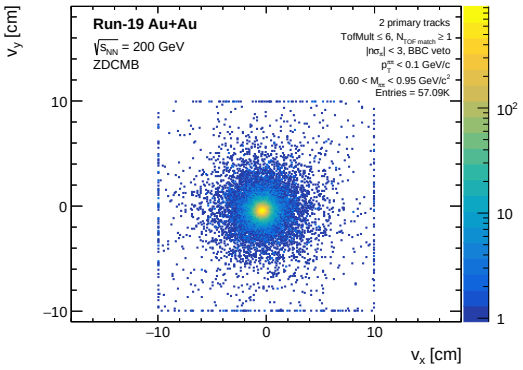


Figure 66: V_x versus V_y distribution of coherent $\pi^+\pi^-$ pairs.

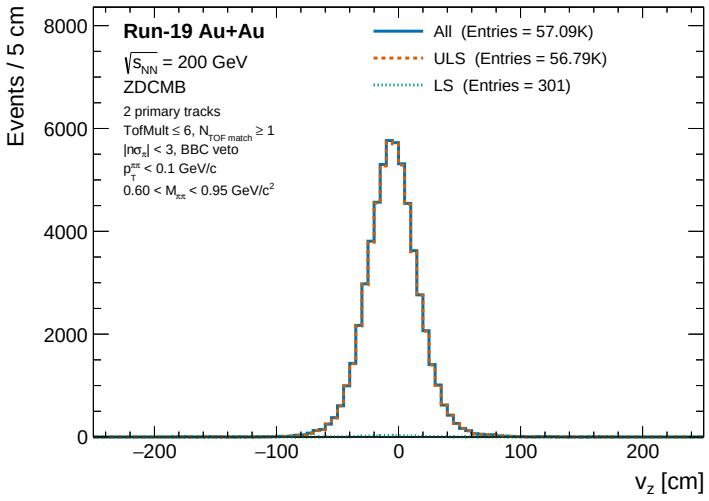


Figure 67: V_z distribution of coherent $\pi^+\pi^-$ pairs.

The V_z distributions are fitted with the same double-Gaussian model as for the UPC-like events, for unlike-sign pairs and for unlike-sign minus like-sign pairs, with and without the $V_r < 3$ cm requirement (Table 11). The fits for unlike-sign pairs without a V_r requirement

and for unlike-sign minus like-sign pairs with $V_r < 3$ cm are shown in Figure 68 and Figure 69.

Table 11: Double-Gaussian fit results for coherent $\pi^+\pi^-$ pairs. In all cases $f_{\text{narrow}} = 1.0000$. All fractions refer to $|V_z| \leq 100$ cm.

Sample	V_r cut	σ_1 [cm]	σ_2 [cm]	w_2	χ^2/ndf	f_{both}	f_{count}
ULS	none	18.6	33.3	0.155	81/57	0.9996	0.9990
ULS − LS	none	18.5	32.4	0.169	79/60	0.9996	0.9993
ULS	< 3 cm	18.5	32.2	0.168	56/48	0.9997	0.9995
ULS − LS	< 3 cm	18.5	32.0	0.172	56/48	0.9997	0.9995

In contrast to the UPC-like events, the wide Gaussian is the same in all four cases ($\sigma_2 \approx$ 32–33 cm, $w_2 \approx 0.16$–0.17): it does not depend on the like-sign subtraction or on the V_r requirement and is therefore part of the luminous region rather than background. Both Gaussians are thus included, and the fits describe the data well ($\chi^2/\text{ndf} \approx 1.2$–$1.4$). The wide Gaussian lies almost entirely inside the fiducial region, so f_{both}, f_{narrow} and f_{count} agree within 0.001. The vertex selection efficiency from the di-pion sample is $\varepsilon_{V_z} = 0.9996$, with a spread of 0.0006 among the variations in Table 11, consistent with the narrow-Gaussian result from the UPC-like events.

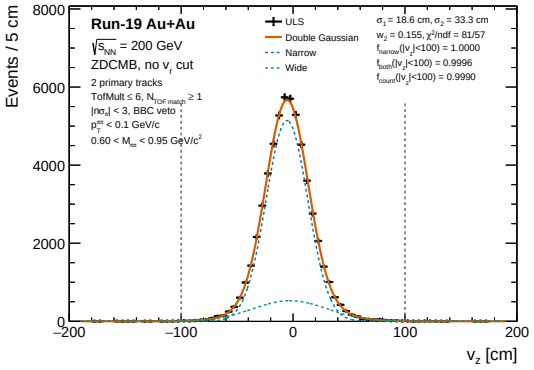


Figure 68: Double-Gaussian fit to the V_z distribution of coherent ULS $\pi^+\pi^-$ pairs without a V_r requirement.

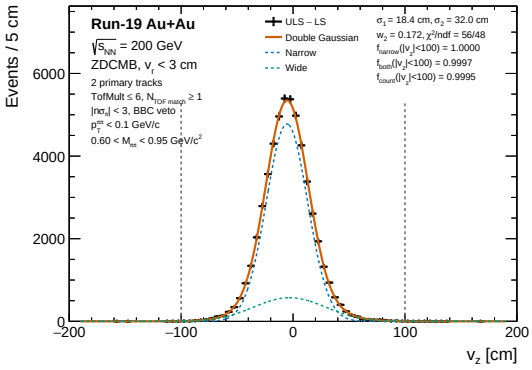


Figure 69: Double-Gaussian fit to the ULS − LS V_z distribution of coherent $\pi^+\pi^-$ pairs with $V_r < 3$ cm.

6.1.3 Nominal Vertex Selection Efficiency

Both methods give a vertex selection efficiency consistent with unity: the narrow Gaussian of the UPC-like events gives $f_{\text{narrow}} = 1.000$, and the coherent $\pi^+\pi^-$ sample gives $\varepsilon_{V_z} = 0.9996$, with all variations within 0.001 of unity. We therefore use $\varepsilon_{V_z} = 1.00$ as the nominal value.

6.2 BBC Veto Selection Efficiency

The BBC veto (Equation 9) rejects events with activity in the small BBC tiles. Besides hadronic background, it also removes genuine UPC events that are accompanied by additional BBC activity, either from the same collision (e.g. electromagnetic dissociation of the nuclei) or from pileup. The corresponding efficiency is measured with coherently photoproduced $\pi^+\pi^-$ pairs, as the ratio of the ρ^0 yields with and without the veto.

The di-pion selection of the vertex study is used, with two changes: the `TofMult` ≤ 6 requirement is removed, since TOF activity is correlated with BBC activity and such a cut would bias the veto efficiency, and no V_z requirement is applied. The selection is therefore exactly two primary tracks on one vertex, at least one track matched to the TOF, $|n\sigma_\pi| < 3$ for both tracks and $p_T^{\pi\pi} < 0.1$ GeV/c as a proxy for coherent selection.

The efficiency can only be measured on a sample that remains signal dominated without the BBC veto. This is ensured by the selection itself. Exactly two primary tracks identified as pions are required, and $\pi^+\pi^-$ photoproduction (dominantly ρ^0) is the most abundant exclusive final state in UPC. In addition, $p_T^{\pi\pi} < 0.1$ GeV/c restricts the sample to the coherent peak, a narrow window in pair transverse momentum where hadronic and incoherent contributions are small. As a result, even without the BBC veto the like-sign pairs amount to only $N_{\mathrm{LS}}/N_{\mathrm{ULS}} = 1.0\%$ in $0.6 < M_{\pi\pi} < 1.2$ GeV/c^2, compared with 0.64% with the veto.

The unlike-sign minus like-sign mass spectra of all selected events, of the events passing the BBC veto, and of the events rejected by it are fitted over $0.6 < M_{\pi\pi} < 1.2$ GeV/c^2 with the modified Söding parametrization (Equation 22), using the same binning and fit procedure as in the vertex study (Figure 70, Figure 71, Figure 72). All three fits describe the data well and their shape parameters agree within uncertainties (Table 12), so the veto changes the yield but not the shape of the spectrum.

None of the samples is background dominated. The remnant background f_p amounts to 5.8% of the unlike-sign minus like-sign pairs for all events and 4.8% for events passing the veto. The slightly larger background without the veto is expected, since the veto preferentially removes hadronic background: only about half of the like-sign pairs (750 to 379) survive the veto, compared with 82% of the signal. Two yields are extracted from each fit: the integral of the fit without the remnant background f_p over the fit range, and the integral of the ρ^0 Breit-Wigner from $2m_\pi$ to $M_\rho + 5\Gamma_\rho$.

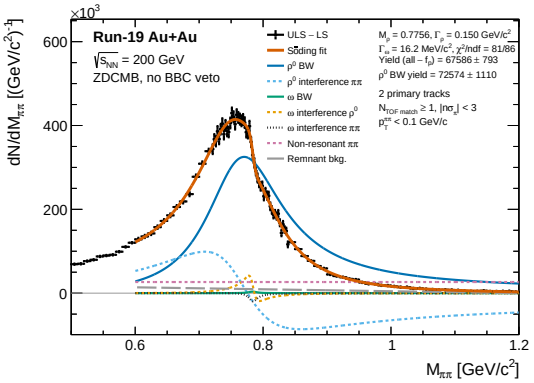


Figure 70: Modified Söding fit to the ULS − LS $\pi^+\pi^-$ spectrum without the BBC veto.

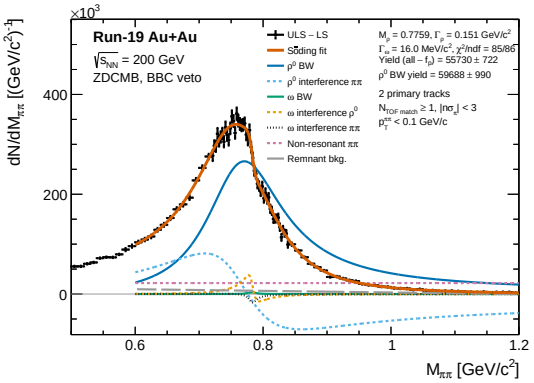


Figure 71: Modified Söding fit to the ULS − LS $\pi^+\pi^-$ spectrum with the BBC veto.

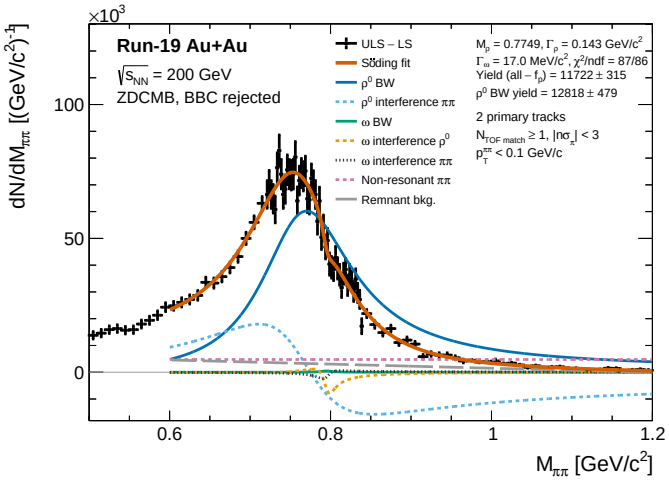


Figure 72: Modified Söding fit to the ULS − LS $\pi^+\pi^-$ spectrum of the events rejected by the BBC veto.

Table 12: Modified Söding fits for all events, events passing the BBC veto, and events rejected by it. Statistical uncertainties from the fits; those of the amplitude ratios include the correlation with A_ρ.

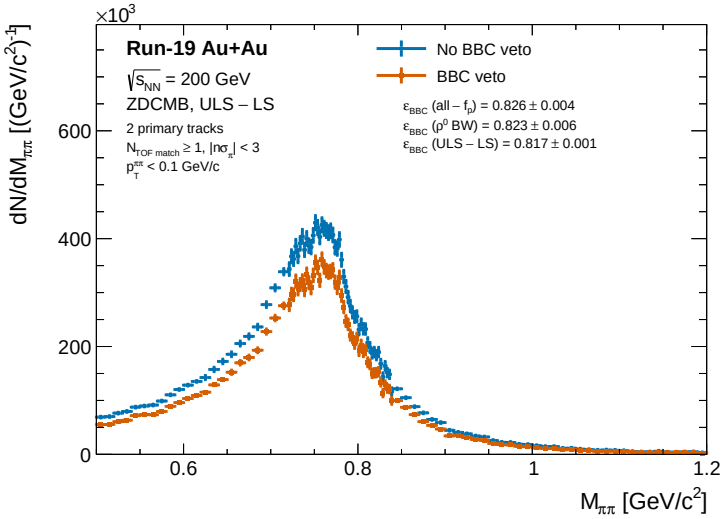
	All events	Pass BBC veto	Rejected by BBC veto
M_ρ [GeV/c^2]	0.7756 ± 0.0012	0.7759 ± 0.0013	0.7749 ± 0.0025
Γ_ρ [GeV/c^2]	0.150 ± 0.003	0.151 ± 0.004	0.143 ± 0.006
$B_{\pi\pi}/A_\rho$ [$(\mathrm{GeV}/c^2)^{-1/2}$]	-0.740 ± 0.017	-0.742 ± 0.018	-0.744 ± 0.040
C_ω/A_ρ	0.283 ± 0.049	0.295 ± 0.056	0.249 ± 0.103
Γ_ω [MeV/c^2]	16.2 ± 4.1	16.0 ± 4.5	17.0 ± 8.2
χ^2/ndf	81/86	85/86	87/86
Yield (all $- f_p$)	67586 ± 793	55730 ± 722	11722 ± 315
ρ^0 BW yield	72574 ± 1110	59688 ± 990	12818 ± 479

The fit to the rejected events (Figure 72) shows that these are predominantly genuine UPC-like di-pions rather than hadronic background. Their spectrum shows the same ρ^0 lineshape, with M_ρ, Γ_ρ and $B_{\pi\pi}/A_\rho$ consistent with those of the accepted events and the ρ^0–ω interference visible at $M_{\pi\pi} \approx 0.78$ GeV/c^2. The background in this sample is larger but still small: $N_{\mathrm{LS}}/N_{\mathrm{ULS}} = 2.7\%$ and f_p amounts to 10% of the unlike-sign minus like-sign pairs. The BBC veto therefore removes genuine UPC signal.

The efficiency is computed from the two disjoint samples as $\varepsilon_{\mathrm{BBC}} = Y_{\mathrm{pass}}/(Y_{\mathrm{pass}} + Y_{\mathrm{rejected}})$. Since the samples do not overlap, their fit uncertainties are independent and are propagated directly; a binomial uncertainty on $Y_{\mathrm{pass}}/Y_{\mathrm{all}}$ would neglect the uncertainty of the fit decomposition, which dominates the yield uncertainties. The efficiency is

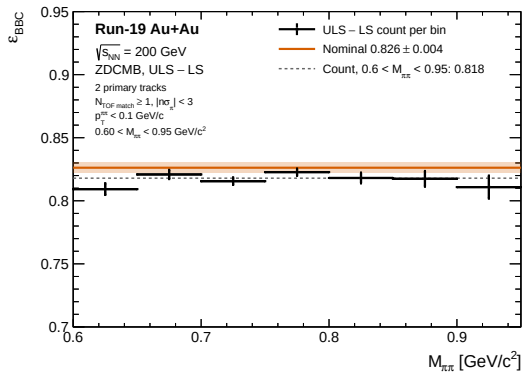
- $\varepsilon_{\mathrm{BBC}} = 0.826 \pm 0.004$ from the yield without f_p,
- $\varepsilon_{\mathrm{BBC}} = 0.823 \pm 0.006$ from the ρ^0 Breit-Wigner yield,
- $\varepsilon_{\mathrm{BBC}} = 0.817 \pm 0.002$ from a direct unlike-sign minus like-sign count over the fit range.

The sum of the passing and rejected yields reproduces the yield of all events to 0.2%, and the ratio $Y_{\mathrm{pass}}/Y_{\mathrm{all}}$ gives consistent values (0.825 and 0.822 for the two fit-based yields). The direct count is lower because it includes the remnant background, which is preferentially removed by the veto. The spectra with and without the veto are compared in Figure 73.



735 Figure 73: Unlike-sign minus like-sign $\pi^+\pi^-$ spectra with and without the BBC veto, with
736 the resulting efficiencies.

737 To check whether the efficiency depends on the kinematics, it is computed in bins of
738 $M_{\pi\pi}$ and of the pair rapidity $y_{\pi\pi}$ within the ρ^0 window $0.60 < M_{\pi\pi} < 0.95$ GeV/c^2, from
739 unlike-sign minus like-sign counts with binomial uncertainties (Figure 74, Figure 75). The
740 efficiency does not vary with the kinematics: compared with the efficiency of the whole
741 window, the bins give $\chi^2 = 8.4$ for 7 bins in $M_{\pi\pi}$ (0.809–0.823) and $\chi^2 = 2.4$ for 8 bins
742 in $y_{\pi\pi}$ (0.814–0.821). The binned counts lie about 0.008 below the nominal value because,
743 like the integrated count, they include the remnant background that the veto removes
744 preferentially. The pair transverse momentum is not used for this check. The selection is
745 restricted to the coherent region, where the sample is clean; towards larger $p_T^{\pi\pi}$ the hadronic
746 background grows, so a dependence on $p_T^{\pi\pi}$ would reflect the changing background fraction
747 rather than the efficiency for the signal.



748 Figure 74: BBC veto efficiency versus $M_{\pi\pi}$
749 from ULS − LS counts.

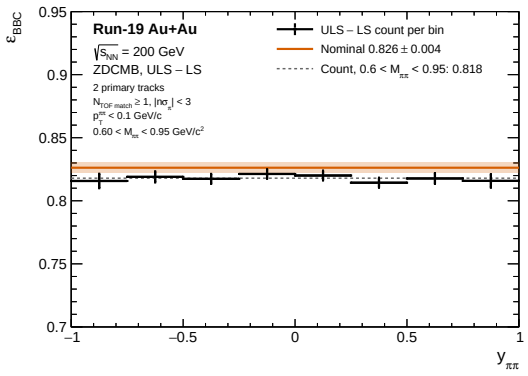


Figure 75: BBC veto efficiency versus the pair rapidity $y_{\pi\pi}$ from ULS − LS counts.

6.2.1 Nominal BBC Veto Efficiency

We use $\varepsilon_{\mathrm{BBC}} = 0.826 \pm 0.004$ as the nominal value, from the yield without f_p, where the uncertainty is statistical, from the fits. The ρ^0 Breit-Wigner yield gives a consistent result.

A Track Distributions of the Selected Candidates

This appendix shows the tracks of the mass-window pairs after the nominal coherent and incoherent selections of Section 5, with both tracks of each pair entered. The distributions are raw reconstructed counts; no requirement beyond the nominal selections is applied.

A.1 Coherent Candidates

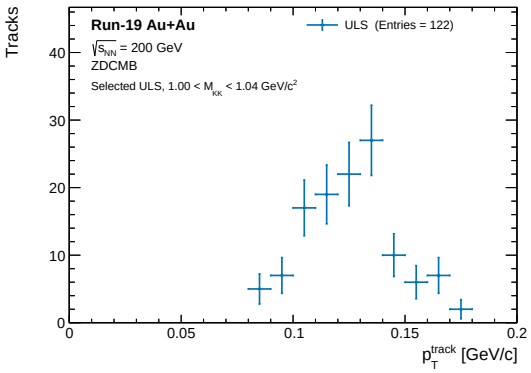


Figure 76: Track p_T of the coherent candidates.

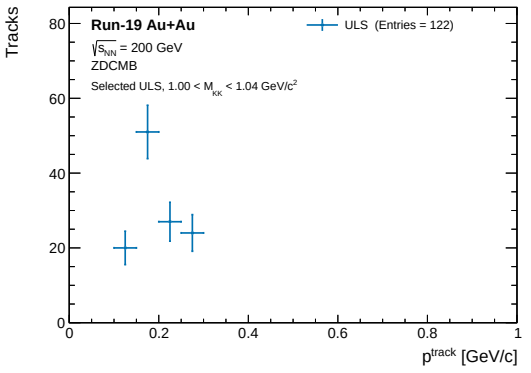


Figure 77: Track momentum of the coherent candidates.

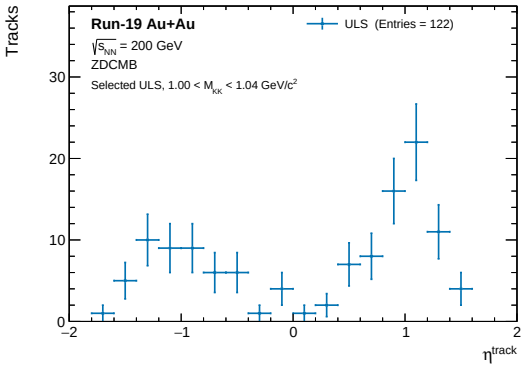


Figure 78: Track pseudorapidity of the coherent candidates.

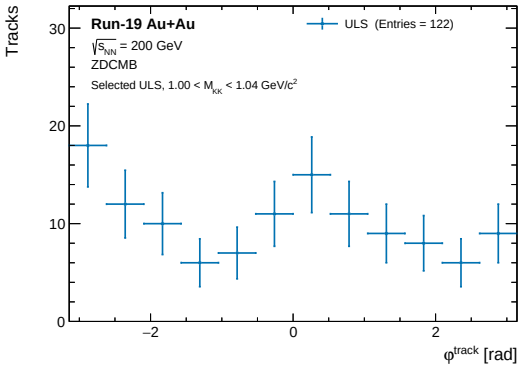


Figure 79: Track azimuth of the coherent candidates.

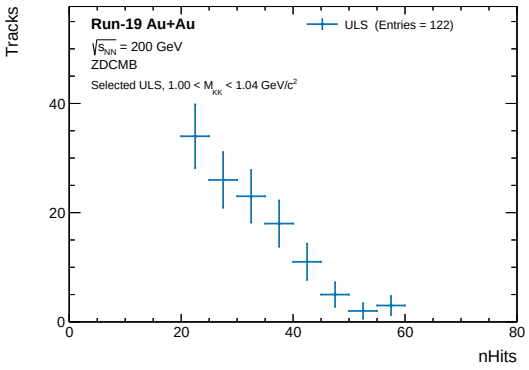


Figure 80: N_{hits} of the coherent candidates.

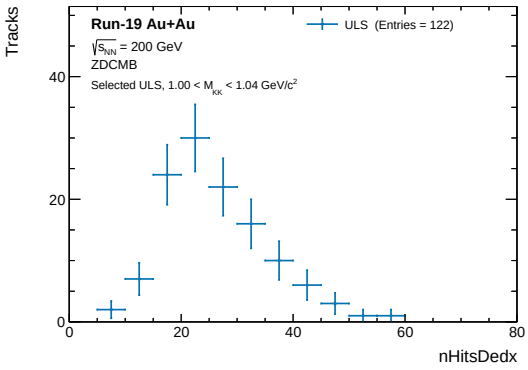


Figure 81: Number of hits used for $\mathrm{d}E/\,\mathrm{d}x$, coherent candidates.

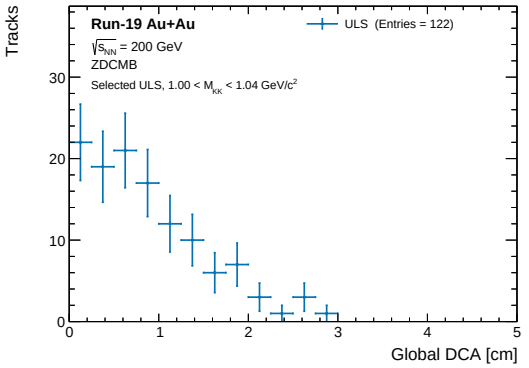


Figure 82: Magnitude of the global-track DCA, coherent candidates.

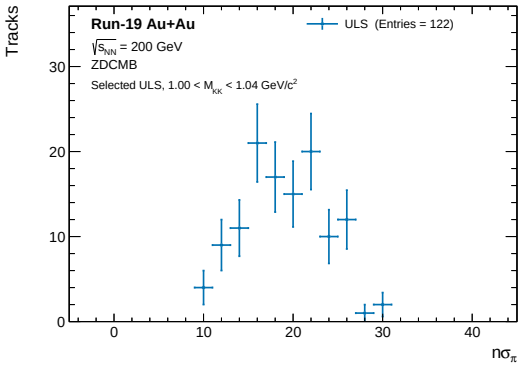


Figure 83: $n\sigma_{\pi}$ of the coherent candidates.

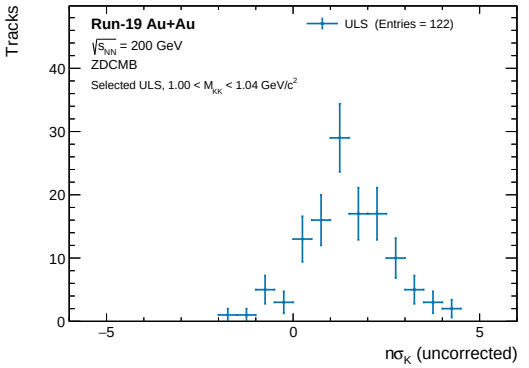


Figure 84: Uncorrected $n\sigma_K$ of the coherent candidates.

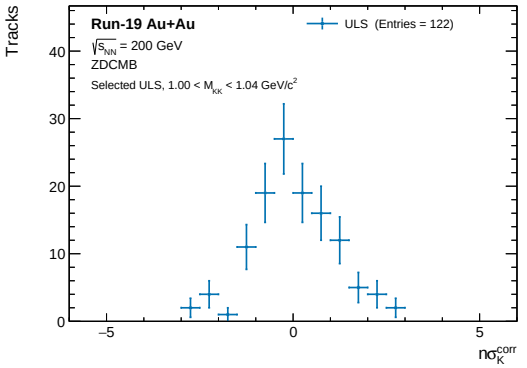
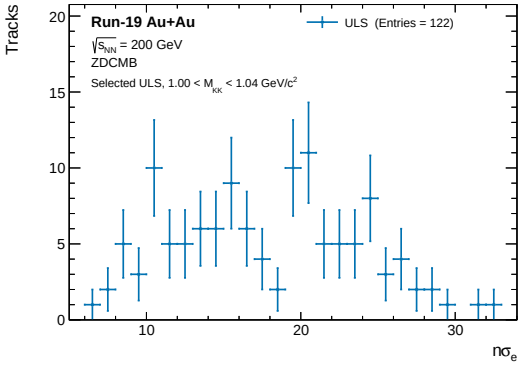


Figure 85: $n\sigma_K^{\mathrm{corr}}$ of the coherent candidates.



768 Figure 86: $n\sigma_e$ of the coherent candidates.

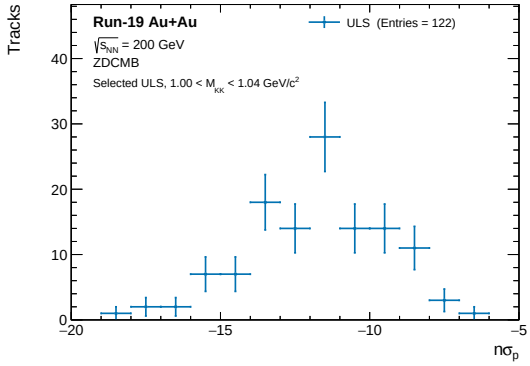
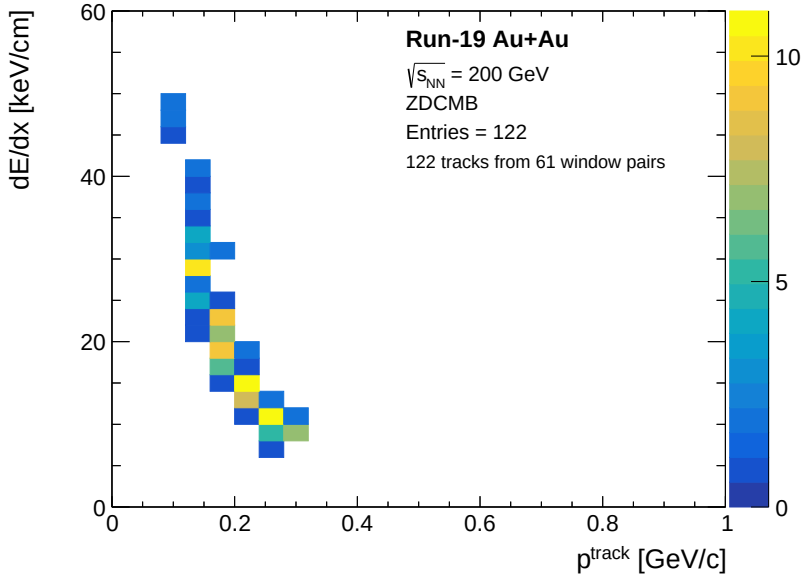


Figure 87: $n\sigma_p$ of the coherent candidates.



769 Figure 88: $\mathrm{d}E/\mathrm{d}x$ versus momentum for the tracks of the coherent candidates.

A.2 Incoherent Candidates

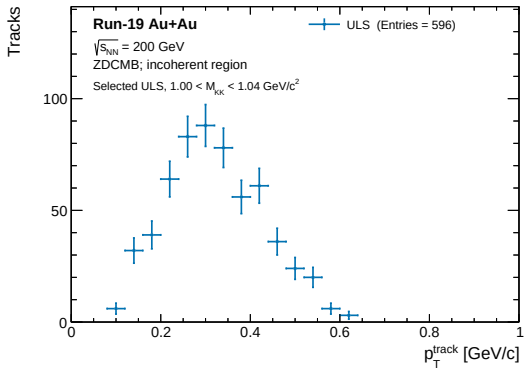


Figure 89: Track p_T of the incoherent candidates.

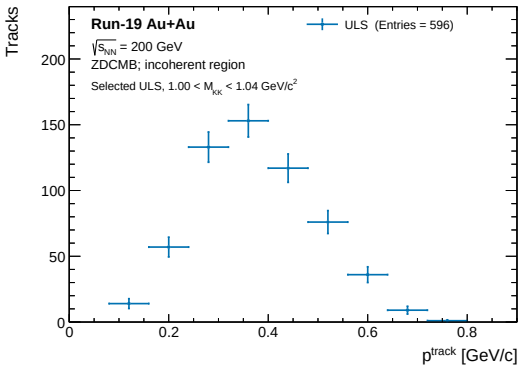


Figure 90: Track momentum of the incoherent candidates.

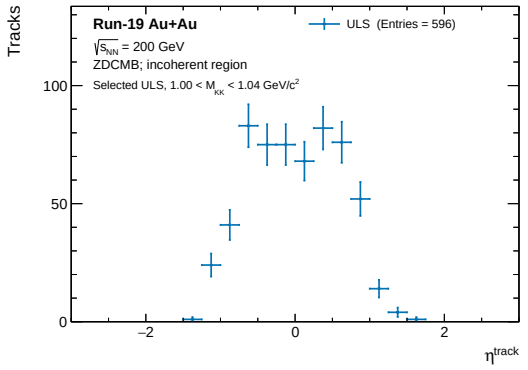


Figure 91: Track pseudorapidity of the incoherent candidates.

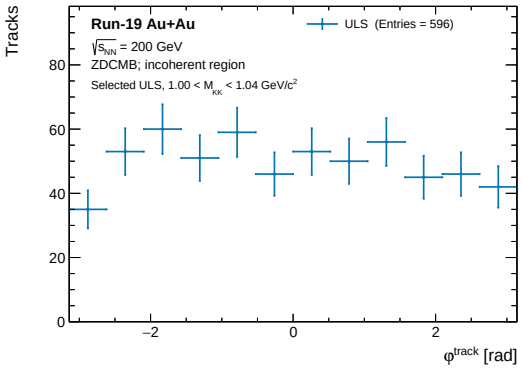


Figure 92: Track azimuth of the incoherent candidates.

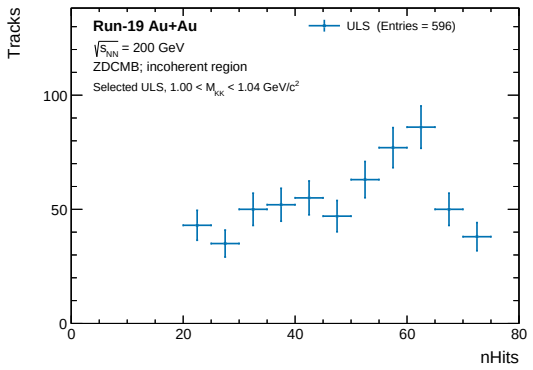


Figure 93: N_{hits} of the incoherent candidates.

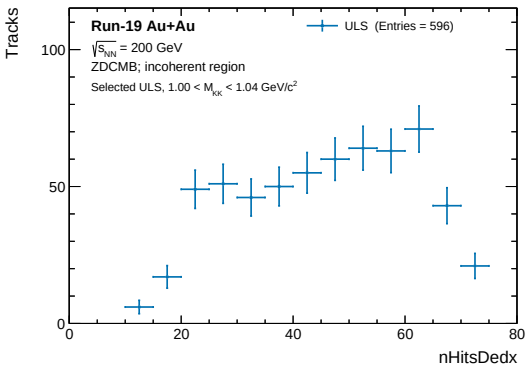
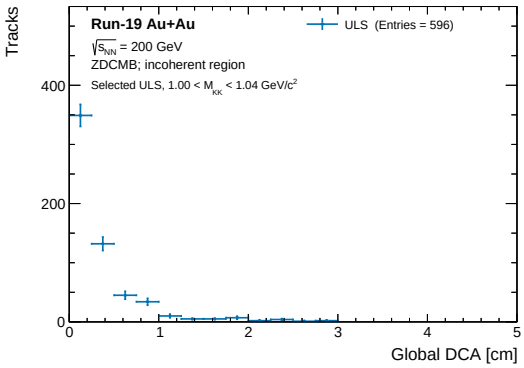


Figure 94: Number of hits used for $\mathrm{d}E/\mathrm{d}x$, incoherent candidates.



777 Figure 95: Magnitude of the global-track
778 DCA, incoherent candidates.

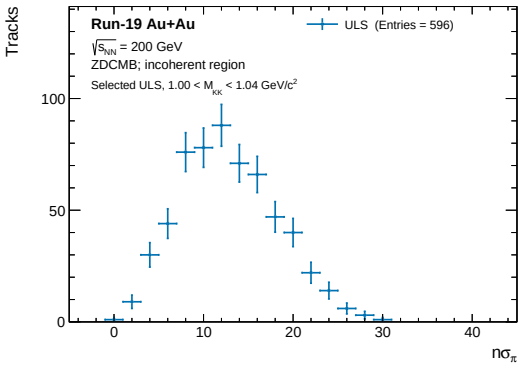
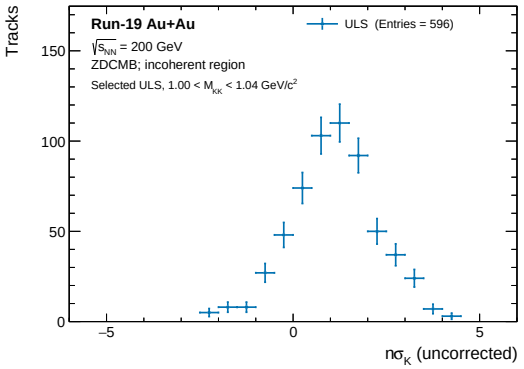


Figure 96: $n\sigma_\pi$ of the incoherent candidates.



779 Figure 97: Uncorrected $n\sigma_K$ of the incoher-
780 ent candidates.

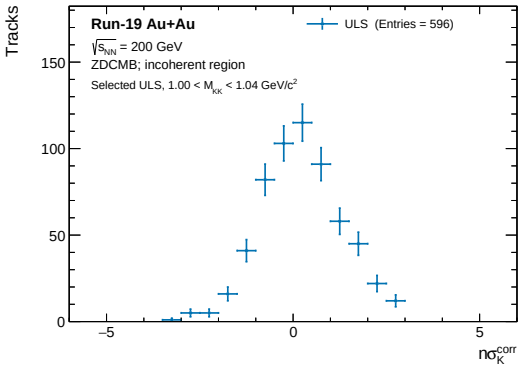
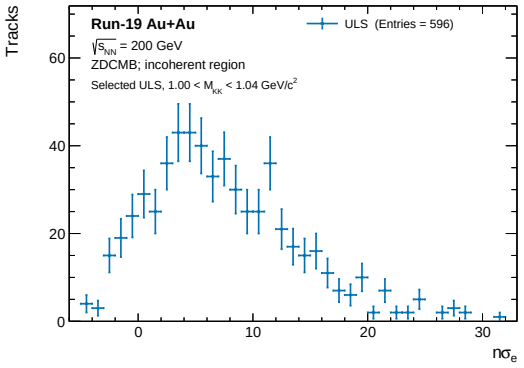


Figure 98: $n\sigma_K^{\mathrm{corr}}$ of the incoherent candidates.



781 Figure 99: $n\sigma_e$ of the incoherent candidates.

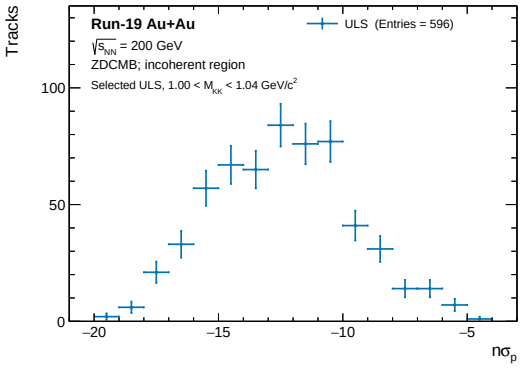
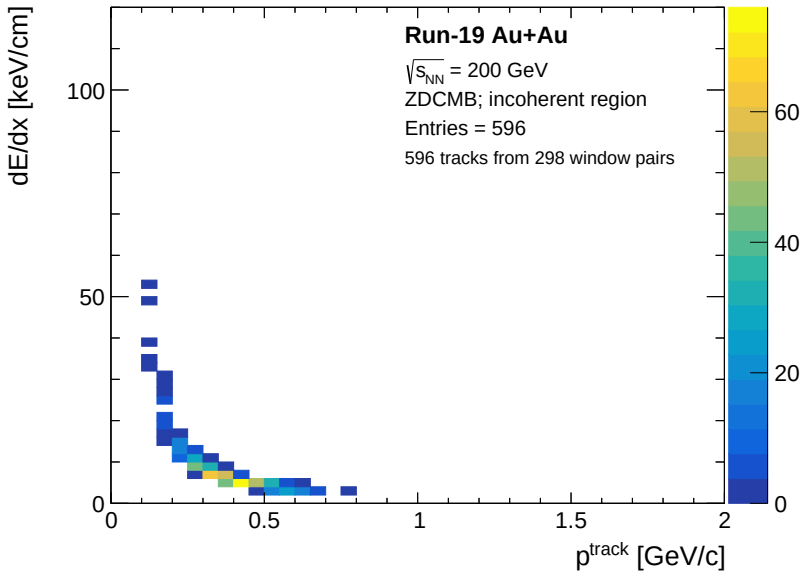


Figure 100: $n\sigma_p$ of the incoherent candidates.

782



783 Figure 101: $\mathrm{d}E/\mathrm{d}x$ versus momentum for the tracks of the incoherent candidates.